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Retail forecasting: Research and practice

Robert Fildes [a,*], Shaohui Ma [b,*], Stephan Kolassa [c]

[a] Lancaster Centre for Marketing Analytics and Forecasting, Department of Management Science, Lancaster University, LA1 4YX, United Kingdom of Great Britain and Northern Ireland
[b] School of Business, Nanjing Audit University, Nanjing, 211815, China
[c] SAP Switzerland, Bahnstrasse 1, 8274 Tägerwilen, Switzerland

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ABSTRACT

This paper reviews the research literature on forecasting retail demand. We begin by introducing the forecasting problems that retailers face, from the strategic to the operational, as sales are aggregated over products to stores and to the company overall. Aggregated forecasting supports strategic decisions on location. Product-level forecasts usually relate to operational decisions at the store level. The factors that influence demand, and in particular promotional information, add considerable complexity, so that forecasters potentially face the dimensionality problem of too many variables and too little data. The paper goes on to evaluate evidence on comparative forecasting accuracy. Although causal models outperform simple benchmarks, adequate evidence on machine learning methods has not yet accumulated. Methods for forecasting new products are examined separately, with little evidence being found on the effectiveness of the various approaches. The paper concludes by describing company forecasting practices, offering conclusions as to both research gaps and barriers to improved practice.

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1. Introduction

The retail industry is experiencing rapid developments in both structure, with the growth in online business, and the competitive environment which companies are facing. There is no simple story that transcends national boundaries, with consumers in different countries behaving in very different ways. For example, in 2017, online retailing accounted for 14.8% of retail sales in the US and 17.6% in the UK, but only 3.4% in Italy – in contrast to Germany, which showed a 3.5% increase to 15.1% since 2015 (Fig. 1).

Retailers' future planning depends in part on demand forecasts, delivered through methods and processes that are embedded in a forecasting support system (FSS). High-accuracy demand forecasting has an impact on organizational performance because it improves many processes along the retail supply chain. In low-margin, high-volume retailing, there is a large and direct improvement in profitability (Fisher & Raman, 2018). At the organizational level, forecasts are essential inputs to many decision activities in functional areas such as marketing, sales, and production/purchasing, as well as finance and accounting. Forecasts also provide the basis for national, regional and local distribution and replenishment plans.

A considerable amount of effort has been devoted to the development and improvement of forecasting models over the past several decades, and retailing has seen a shift from intuition to data-based decision making that can capitalize on this (Fisher & Raman, 2018). This paper reviews the research as it applies to retail forecasting, drawing boundaries around the field to focus on food and non-food including electrical goods (but excluding for example cars, petrol or telephony). This broadly matches the definitions and categories adopted in the UK and US government retail statistics, for example. We give special

* Corresponding authors.
E-mail addresses: R.Fildes@lancaster.ac.uk (R. Fildes), shaohui.ma@nau.edu.cn (S. Ma), stephan.kolassa@sap.com (S. Kolassa).

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consideration to internet sales, as there is less and less of a distinction between internet sales and brick-and-mortar retail. We exclude mail order and catalog sales – although online retailers may be able to learn from this field of study, the differences are fundamental (online retailers can use dynamic promotions, product recommendations tailored to a customer's search and purchase history, and availability-to-promise, including recommending substitutes) and the free-standing catalog business is quite obviously dying out. Our objective is to draw together and critically evaluate a diverse research literature in the context of the practical decisions that retailers must make that depend on quantitative forecasts. This examination looks at the variety of demand patterns in the different marketing (especially promotional) contexts and levels of aggregation at which forecasts must be made in order to support decisions, from the strategic to the operational.

We relied mainly on Scopus for literature searching. As retail forecasting is an extensive topic and concerns a wide range of research fields, many papers do not use 'retail*' as the keyword, but instead use more specific ones, such as 'promotion', 'supply chain', 'store', 'fashion', 'product', or 'demand/sales'. This makes the collection of retail forecasting papers for review a difficult task. To ensure a comprehensive review, we built a number of keyword compositions, e.g. "(retail or demand or sales) and (forecasting or prediction)", for the literature search, then browsed each of the papers thus identified to judge the extent of their relevance to retail forecasting.

The next section considers the decisions that retailers make, from the strategic to the operational, and the different levels of aggregation, from the store up to the retail chain. Section 3 considers aggregate forecasting from the market as a whole, where, as we have noted, rapid changes are taking place, down to the individual store, where again the question of where stores should be located has risen to prominence with the changes that have been observed in shopping behavior. We next turn to more detailed stock keeping unit (SKU) forecasting, and the hierarchies that these SKUs naturally fall into, discussed in Section 4. The data issues that are faced when forecasting include stock-outs, seasonality and calendar events, while the key demand drivers are the marketing mix and promotions. Online product reviews and social media are two new information sources that require considerable care if they are to prove valuable for forecasting. Section 5 provides an evaluation of the different methods that are used in product-level demand forecasting in an attempt to provide evidence as to the circumstances in which more complex methods add value, including a short discussion of the specific challenges to forecast accuracy evaluation at a low granularity. New product forecasting requires different approaches, which are considered in Section 6. Forecasting practice varies dramatically across the retail sector, in part because of its diversity, and Section 7 provides various vignettes based on case observations, which capture some of the issues that retailers face and their operational solutions. Finally, Section 8 contains our conclusions as to both those areas in which the evidence as to best practice is strong and those in which research is most needed.

2. Retailers' forecasting needs

Retailers rely on forecasts to support strategic, tactical and operational decisions. Each level has a different focus; nevertheless, there is a need for compatibility across the three levels. Methods for ensuring this are considered in Section 4.1.

2.1. Strategic level

Retailers, like all organizations, must make decisions regarding their strategic development within changing competitive and technological environments. The standard elements that define a retail strategy, embracing market and competitive factors within the developing technological and regulatory environments (see for example Levy, Weitz, & Grewal, 2012), typically depend on forecasts. Fig. 1 illustrates these issues, showing the recent and forecasted growth of online purchases in the US and Europe, with clear differences between countries and cultures. Fig. 2 shows the changing share of low-price retailers in the UK from 1994 to 2017, with forecasts to 2020.

The simple extrapolative forecasts in Figs. 1 and 2 show the rapid changes in the retail environment that require companies to respond. For example, a channel decision to develop an online presence will depend on a forecast time horizon that looks years ahead, but with some quantitative precision being required over shorter horizons, perhaps as soon as a possible implementation a year or more ahead. The retailer chain's chosen strategy will require decisions that respond to the above changes, deciding on locations including channels, price/quality position and target market segment(s), store type (in town vs. megastores) and distribution network. A key point is that typically, all such decisions will have long-term consequences, with high costs being incurred if subsequent changes are needed and flexibility being low (e.g., site location and the move to more frequent local shopping in the UK, away from the large out-of-town stores, leading Tesco in 2015 to sell 14 of its earmarked sites in the UK and close down others and Marks & Spencer Group in 2018 to propose to close down more than 10% of its stores). Thus, strategic forecasts are required at both a highly aggregate level and a geographically-specific level over long forecast horizons.

The small local retailer faces an equally volatile environment, with uncertainty as to the location and target market (and product mix). However, while many of the questions faced by the national retailers remain relevant (e.g. online offering), there is little in the research literature that even describes the results of the many small shop location decisions. Exceptions include charity shops (Alexander, Cryer, & Wood, 2008) and convenience stores (Wood & Browne, 2007), while a number of studies have also examined restaurants, which are outside our scope.

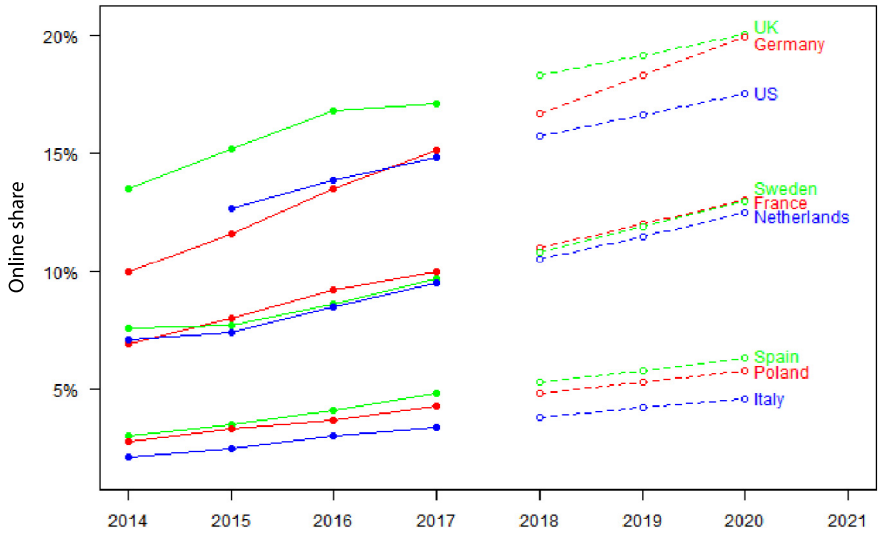


Fig. 1. Online shares of retail trade: Various countries. Source for the 2014–2017 data: Center for Retail Research, www.retailresearch.org/onlineretailing.php. The forecasts for 2018–2020 were calculated using exponential smoothing with additive trends and errors

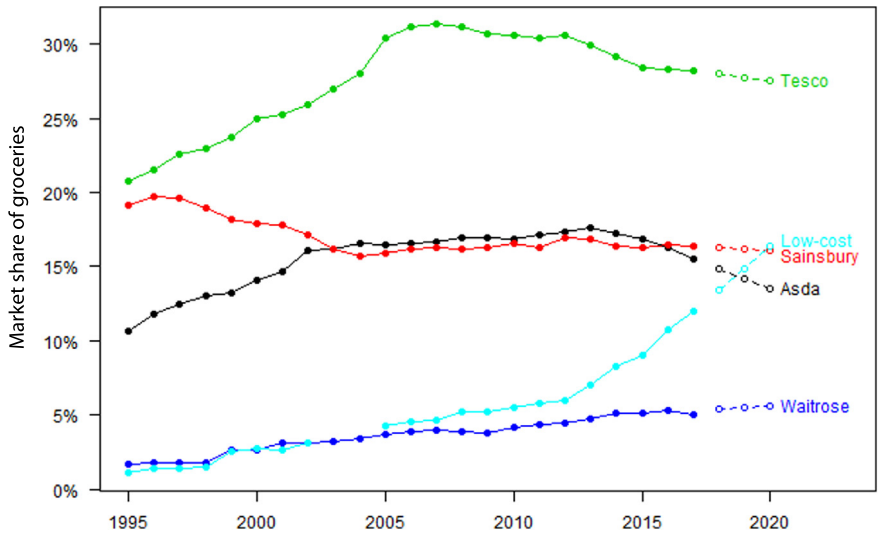


Fig. 2. Shares of grocery retailers compared to low price retailers (Aldi and Lidl) in the UK. Source for the 1995–2017 data: http://www.fooddeserts.org/images/supshare.htm. The forecasts for 2018–2020 were calculated using exponential smoothing with additive trends and errors.

2.2. Tactical level

Tactical decisions necessarily fit within the strategic framework developed above. However, these strategic decisions do not determine the communications and advertising plan for the chain, the categories of products to be offered, or the variety (range) of products within each category. At the chain level, the aim is to maximize the overall profitability by using both advertising (at the chain and store levels) and promotional tools to achieve success.

At the category level, the objective again is to maximize category (rather than brand) profits, which will require a pricing/promotional plan that determines aspects such as the type and frequency of promotions over the planning horizon (of perhaps a year), and whether there are associated display and feature advertising campaigns. In principle, these plans are linked to the operational promotional pricing decisions discussed below. The on-shelf availability of products is also a key metric of retail service, and depends crucially on establishing a relationship

between the product demand forecasts, inventory investment and the distribution system. The range of products listed raises the question of new product introduction into a category, the expected sales, and its effect on sales overall (particularly within the category).

The demands placed on the warehouse and distribution system by the store × product demand also need forecasting. This is necessary in order to plan not only replenishment, but also the workforce, where the number and size of products determine the pick rate, which in turn determines the workforce and its schedule. The constitution of the delivery fleet and planned routes similarly depend on store demand forecasts (somewhat disaggregated), since seasonal patterns of purchasing vary by region. This is true whether the retailer runs its own distribution network or has outsourced it to a service provider – or, which is most common, uses a mixture, with many products being supplied from the retailer's own distribution centers (DCs), but others being supplied directly to stores by the manufacturers (direct store delivery). Online retailers, especially grocers, need to decide on what mode of fulfillment to offer: store or curbside pickup, home delivery out of stores or DCs (Glaeser, Fisher, & Su, 2019; Hübner, Kuhn, & Wollenburg, 2016), each of which requires specific forecasts for demand and workforce requirements.

2.3. Operational level

To be successful in strategic and tactical decisions, the retail company needs to manage its demand and supply planning processes so as simultaneously to avoid customer service issues, unnecessarily high inventory and substantial write-off costs due to obsolete products. These are sensitive issues in retail companies because of the complexity in the demand data, with considerable fluctuations, the presence of many intermediaries in the process, the diversity of products and the service quality required by the consumer. In a general way, accurate demand forecasting is crucial in organizing and planning purchasing, distribution, and the labor force, as well as after-sales services. Thus, retail managers' ability to estimate the expected sales quantity at the SKU × store level over the short term should lead to improved customer satisfaction, reduced waste, increased sales revenue and a more effective and efficient distribution. However, point forecasts need to be combined with estimates of quantiles (or the standard deviation of the forecast error over the forecast horizon) if they are to be effective.

As a result of these various operational decisions and their respective financial consequences, the cash that retailers generate (since suppliers are usually paid in arrears) leads to a cash management investment problem. Thus, the cash that will be available for investment, which itself depends on the customer payment arrangements, needs to be forecast.

Finally, day-to-day store operations are also forecast-dependent. In particular, staffing schedules depend on the anticipated customer activity and product intake.

3. Aggregate retail sales forecasting

All forecasting in retail depends on a degree of aggregation over product units, locations or time buckets, according to the objective of the forecasting activity.

This section uses the term 'aggregate retail sales forecasting' to refer to the total retail sales in a market, chain, store type (e.g. in-town) or individual store, as opposed to product (SKU/brand/category) specific forecasts; i.e., we implicitly aggregate across products and promotions and up to a specific granularity (e.g. weekly or monthly) over a specific time period, see Fig. 3. Usually, aggregate retail sales are measured in revenue instead of product units. However, price increases for inflation usually have little impact on the demand, so the effect on the total revenue may be captured more easily by forecasting in units and multiplying by price. Below, we review the existing research on three separate levels: the aggregate retail sales in a market, a chain, and a store. Although the forecasting of aggregate sales at these three levels shares many common issues, e.g. seasonality and trend, they raise different forecasting questions and have different objectives, data characteristics, and solutions.

3.1. Market-level aggregate sales forecasting

Market-level aggregate sales forecasting concerns the total sales of a retail category, a channel, or the whole industry in a country or region. The time bucket for the market-level forecasts may be monthly, quarterly or yearly. Forecasts of market-level retail sales are necessary to enable (large) retailers to understand changing market conditions and the effect that these may have on their own total sales (Alon, Qi, & Sadowski, 2001). They are also central to the planning and operation of a retail business at the strategic chain level, in that they help to identify the growth potential of different business models and stimulate the development of new strategies for maintaining their market position.

Market-level aggregate retail sales data often exhibit strong trends, seasonal variations, serial correlation and regime shifts, because any long span in the data may include economic growth, inflation and unexpected events (Fig. 4). Time series models have provided solutions for capturing these characteristics, and thus, they have long been being applied for market-level aggregate retail sales forecasting (e.g., Alon et al., 2001; Bechter & Rutner, 1978; Schmidt, 1979; Zhang & Qi, 2005). Simple exponential smoothing and its extensions, along with ARIMA models, have been the time series models employed most frequently for market-level sales forecasting. Even in the earliest references, reflecting controversies in the macroeconomic literature, researchers raised the question of which of various time series models performed best and how they compared to simple econometric models.[1] Early studies suffered from a common weakness: a failure to compare models convincingly because of a reliance on

[1] Typically, macroeconometric models do not include retail sales as an endogenous variable, but rather use a variable such as consumption; however, see Bechter and Rutner (1978).

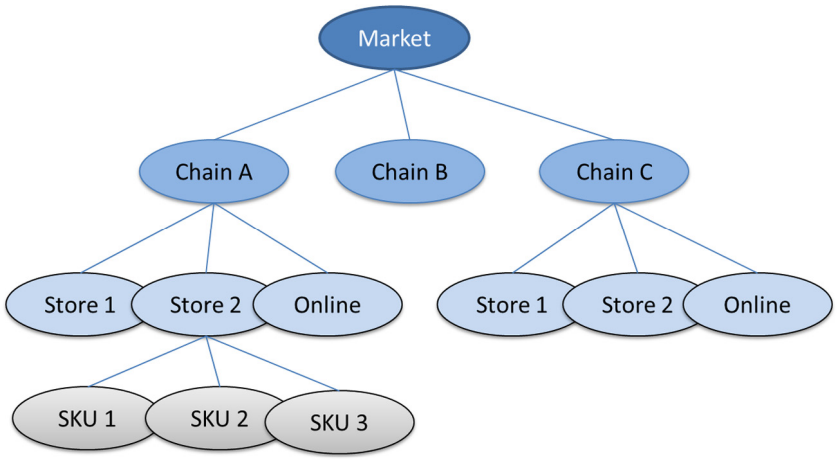


Fig. 3. Hierarchical retail sales from SKU to Store to Chain to Market.

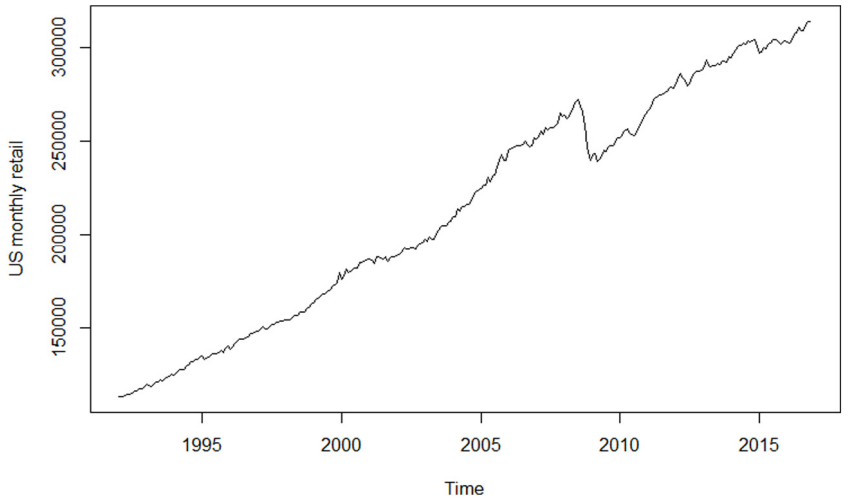


Fig. 4. US retail sales monthly series in millions of dollars.
Source: U.S. Census Bureau.

limited data or a use of inappropriate evaluation metrics such as the in-sample fit.

Some researchers have found that standard time series models are sometimes inadequate for approximating aggregate retail sales, identifying evidence of nonlinearity and volatility in market-level retail sales time series (e.g., Alon et al., 2001; Chu & Zhang, 2003; Kuvulmaz, Usanmaz, & Engin, 2005; Zhang & Qi, 2005). They therefore resorted to nonlinear models, especially artificial neural networks. The results indicate that traditional time series models with a stochastic trend, such as exponential smoothing and ARIMA, performed well when macroeconomic conditions are relatively stable. However, when economic conditions are volatile (with rapid changes in economic conditions), artificial neural networks (ANNs) have been claimed to outperform the linear methods (Alon et al., 2001), though there must be a suspicion of overfitting, since only a single time series was being forecast and the authors presumably tuned their models to improve the forecast accuracy. Evidence for the forecasting benefits of nonlinear models seems weak (see below).

Unlike time series models, econometric models depend on the successful identification of predictable explanatory variables. Bechter and Rutner (1978) compared the forecasting performances of ARIMA and econometric models designed for US retail sales. They used two explanatory variables in the economic model: personal income and nonfinancial personal wealth, as measured by an index of the price of common stocks. Past values of retail sales were also included in alternative models that mixed autoregressive and economic components. They found that ARIMA forecasts were usually no better and

often worse than those generated by a simple single-equation economic model, while the mixed model had a better record over the entire 30-month forecast period than any of the other three models. Note though that they used the actual values of the macroeconomic predictors for the last observed month to forecast one month out, whereas in a practical forecasting situation, these variables would need to be forecasted (see the discussion of forecasting predictors in Section 4.3 on product-level demand drivers). No *ex ante* unconditional forecast comparisons have been found. Recently, Aye, Balcilar, Gupta, and Majumdar (2015) conducted a comprehensive comparative study over 26 (23 single and 3 combination) time series models for forecasting South Africa's aggregate retail sales. Unlike the previous literature on retail sales forecasting, they not only looked at a wide array of linear and nonlinear models, but also generated multi-step-ahead forecasts using a real-time recursive estimation scheme over the out-of-sample period. In addition, they considered loss functions that overweight the forecast error in booms and recessions. They found that no unique model performed best across all scenarios. However, combination forecast models, which weight current information more than past, not only produced better forecasts, but also were largely unaffected by business cycles and time horizons.

In summary, no recent research has been found that uses current econometric methods to link retail sales to macroeconomic variables such as GDP and evaluates their conditional and unconditional performances relative to time series approaches. The evidence on the performance of nonlinear models is limited, with too few series from too few countries, and no comparison with econometric models has been made.

3.2. Chain- and channel-level aggregate sales forecasting

Research at the retail chain level has focused mainly on sales forecasting one year ahead (Curtis, Lundholm, & McVay, 2014; Kesavan, Gaur, & Raman, 2010; Osadchiy, Gaur, & Seshadri, 2013). Accurate forecasts of chain-level retail sales (in monetary terms) are needed for company financial management and also to aid financial investment decisions in the stocks of retail chains.

In general, most of the models that are used for chain-level forecasting are similar to those used for market-level forecasting (i.e., univariate extrapolation models). However, there are some specially-designed models which have been found to have better performances. Kesavan et al. (2010) found that inventory and gross margin data can improve the forecasting of annual sales at the chain level in the context of publicly-quoted U.S. retailers. They incorporated the cost of goods sold, inventory, and gross margin (the ratio of sales revenue to cost of goods sold) as endogenous variables in a simultaneous equations model, and found the sales forecasts from this model to be more accurate than the consensus forecasts from equity analysts. Osadchiy et al. (2013) presented a (highly structured) model for incorporating lagged financial market returns and financial analysts' forecasts when forecasting firm-level sales for retailers. Their method improved upon the accuracy of forecasts generated by both equity analysts and time series methods. However, their use of benchmark methods (in particular a more standard econometric formulation) was limited. Curtis et al. (2014) forecast retail chain sales using publicly-available data on the age mix of stores in a retail chain. By distinguishing between the growth in sales-generating units (i.e., new stores) and the growth in sales per unit (i.e., comparable store growth rates), their forecasts proved significantly more accurate than either those from models based on estimated rates of mean reversion in total sales or analysts' forecasts. Internal models of chain sales forecasts should benefit from the inclusion of additional confidential variables, but no evidence has been found.

A related question in forecasting the potential online demand is deciding whether to add an online channel to existing brick-and-mortar stores – or whether to close down an existing online presence. This analysis needs to allow for the cannibalization of existing locations by online sales. Hernant and Rosengren (2017) studied the effects on customer behavior, both in-store and online, of the launch of an online channel, and concluded that "for existing customers, the interaction between average transaction, purchase frequency, and regularity turned out to be a zero-sum game". More positively, though, the addition of an online service offered the opportunity to gain new customers who had not purchased in-store.

The results as to how online is penetrating brick-and-mortar retail are known to be both category- and region-specific (Brynjolfsson & Rahman, 2009; Wang, Song, & Yang, 2013). Critically, the final online penetration into each category is unknown, and again, the question has not been researched. For a retailer who is contemplating the development of an online service, a hierarchical approach would be needed. However, anecdotal evidence (including one of our case vignettes in Appendix B) suggests that the two channels may be forecast separately.

3.3. Store-level aggregate sales forecasting

Retailers typically have multiple stores of different formats that serve different customer segments in different locations. Store sales are impacted dramatically by location, the local economy and competitive retailers, consumer demographics, own and competitor promotions, weather, seasons, and local events such as festivals. The forecasting of store sales can be divided into two categories: (1) the forecasting of existing store sales for distribution, target setting and viability, financial control, and workforce planning; and (2) the forecasting of new store potential sales for site selection analysis.

Both univariate time series models and regression models are used for forecasting existing store sales. Davies (1973) used factor analysis in a clothing-chain study and demonstrated how the factor scores of individual stores explained their sales performance levels. However, this study was limited by its use of in-sample data only. Geurts and Kelly (1986) presented a case study of forecasting the monthly sales of a department store. They considered various factors, including seasonality, holidays, number of weekend days, the local consumer price

index, average weekly earnings, the unemployment rate, etc., and concluded that univariate time series methods were better for forecasting store sales than judgment or econometric models. At a more operational level of managing staffing levels, Lam, Vandenbosch, and Pearce (1998) built a regression model based on daily data which set the store sales potential as a function of the store traffic volume, customer type, and customer response to sales force availability, where the errors were modelled as ARMA processes. However, no convincing evidence was presented on its comparative accuracy.

A related store problem is that of forecasting activity or footfalls in a shopping center or individual store, for which previously data collection has been problematic (Abrishami, Kumar, & Nienaber, 2017), but now new technologies provide data. For example, the short-term forecasting of store activity can utilize recently-available 'big' data in the form of third-party mobile payment transactions in order to produce daily shop sales forecasts for 2000 shops at various locations (Ma & Fildes, 2018). The results of such an exercise showed that forecasts which took into account the overall activity on a third-party mobile payment platform (i.e., a pooled multivariate approach), produced using a machine learning algorithm, outperformed univariate methods, including standard benchmarks, both in aggregate and for most individual stores.

The forecasting of a new store's sales potential is a difficult task, but one that is crucial to the success of every retailing company. Traditionally, new store sales forecasting approaches can be classified into three categories: judgmental, analogue regression and spatial interaction models (also called gravitational models). The success of judgmental approaches depends on the experience of the location analyst (Reynolds & Wood, 2010) and, as with all judgmental approaches, the extent to which they can avoid a range of well-documented cognitive and motivational biases. Retailers often use a checklist to assess systematically the relative value of a given site compared to other potential sites in the area. This involves an evaluation of various factors that are likely to have an impact upon the sales and costs at a site (Craig, Ghosh, & Mclafferty, 1984; Hernandez & Bennison, 2000), and can deal with issues that cannot easily be measured quantitatively (e.g. access and visibility). In its simplest form, the checklist can act as a good screening tool, but it is unable to predict turnover directly.

The analogue regression generates turnover forecasts for a new store by comparing the proposed site with analogous existing sites, based on the measurement of features such as competition (number of competitors, distance to key competitor, etc.), trading area composition (population size, average income, the number of households, commute patterns, car ownership, etc.), store accessibility (cost of parking, distance to parking, distance to bus station, etc.) and store characteristics (size, format, brand image, product range, opening hours, etc.). Compared with the judgmental approach, analogue regression models provide a more objective basis for the manager's decision-making, highlighting the most likely options for new locations. Simkin (1989) reported the successful application of a regression-based store location assessment model (SLAM) in several of the UK's major retailers. The model was able to account for approximately 80% of the store turnover in-sample, but prediction accuracies for the sales of new stores were not reported. Morphet (1991) applied regression to an analysis of the trading performance of a chain of grocery stores in England, incorporating five competitive and demographic factors (including population, share of floor space, distance from higher order centre, pull, percentage of married women, etc.). Though the models achieved a high degree of 'explanation' of the variation in store performance, the results on the predicted turnover suggested that the use of regression equations was not sufficient for predicting the potential performances of stores in new locations. The pitfalls in such regressions may come from statistical overfitting due to limited data, neglecting consumer perceptions, an inadequate coverage of the competition, and uncertainty induced by having to forecast explanatory variables themselves. While the method can include various demographic variables, and is therefore appropriate for retail operations that are aiming for a segmented market, it is heavily data dependent and therefore of limited value for a rapidly changing retail environment (as in the UK).

The spatial interaction model (SIM) (or gravity model) is used widely as a sophisticated retail location analysis tool, and has a long and distinguished history in the fields of geography and regional science. Based on Reynolds and Wood (2010) survey of corporate location planning departments, around two thirds of retail location planning teams (across all sectors) had made use of SIM for location planning. Unlike analogous regressions, which mainly rely on data from existing stores in the same chain, SIM uses data from various sources to improve the prediction accuracy: analogous stores, household surveys, geographical information systems, and competition and census data. A SIM is based on the theory that expenditure flows and subsequent store revenue are driven by the store's comparative attractiveness and constrained by distance, with consumers exhibiting a higher likelihood of shopping at stores that are geographically proximate (Newing, Clarke, & Clarke, 2014). The most basic example of this type of model is the Huff trade area model (Huff, 1963). Comparative attractiveness in a SIM will depend on factors such as accessibility, parking, other store features, etc. SIM'S popularity and longevity can be attributed to its conceptual appeal, relative ease of use, and applicability to a broad range of problems, of which the prediction of consumer spatial behavior is the most widely known (Li & Liu, 2012). The original Huff model has been extended to make it more realistic by adding additional components; these include models that can take into account retail chain image (Stanley & Sewall, 1976), asymmetric competition in retail store formats (Benito, Gallego, & Kopalle, 2004), store agglomeration effects (Li & Liu, 2012; Picone, Ridley, & Zandbergen, 2009; Teller & Reutterer, 2008), retail chain internal cannibalization (Beule, Poel, & Weghe, 2014), and consumer heterogeneity (Newing et al., 2014). Recently, Glaeser et al. (2019) provided a solution to the problem of optimally selecting locations for the pickup of purchases made online, using a random forest including geographic and other predictors; this included the use

of service variables for capturing the attractiveness of different locations.

SIMs are usually validated on in-sample data. However, Birkin, Clarke, and Clarke (2010) criticized this approach, emphasizing the importance of a hold-out sample (an unacknowledged reference to the forecasting literature), and showed, using DIY chain store data, that their model could be operationalized with a forecasting accuracy of around 10%, which was better than the company's performance. An important omission is the time horizon over which the model is assumed to apply, presumably the time horizon of the investment. Birkin et al. (2010) comment that industry practice is to update the models regularly, at least annually, which suggests an implicit view regarding the lack of longer-term stability in the models, due to a changing retail environment. Extensions to the model suffer from problems of data inadequacy, but Newing et al. (2014) argue that these can be overcome in order to include more sophisticated demand terms such as seasonal fluctuation and consumer segmentation variables.

Predictive models of store performances are only one element in supporting the location decision. Wood and Reynolds (2013) discuss how the models are combined with context-specific knowledge, the judgments of location analysts and analogous information to produce final recommendations. There is no evidence available on the relative importance of judgmental inputs and model-based information, nor is there much evidence on the accuracy of the models beyond untested claims that the model-based forecasts are highly accurate (Wood & Reynolds, 2013), apart from Birkin et al.'s (2010) analysis of a DIY chain. In the current rapidly changing retail environment, we speculate that judgment will again become the dominant approach to the evaluation of store potential and store closures. The research question now becomes: what role – if any – can models usefully play, and how should the weights of model and judgmental contributions be assigned?

With the rapid changes on the high-street in many countries showing increasing vacancy rates, these forecasting models will increasingly have a new use: for identifying shops to be closed. We speculate that multivariate time series models that include indicator variables (for the store type), supplemented by local knowledge, should prove useful. However, this research remains to be done.

Note that any evaluation of new store forecasts (or of forecasts that might be used to determine which stores to close) needs to take a potential selection bias into account: candidate new stores with higher forecasts are more likely to be developed than candidates with lower forecasts. The analogue holds for stores that are closed. The candidates selected for development may therefore see systematically lower sales than was forecasted, because of regression to the mean. Importantly, this effect may already bias the samples that are available for new store forecasting research: usually, a researcher will be able to rely only on data for stores that were actually developed (meaning that someone had already forecasted their sales, and the forecast came out high enough to go forward with the development); one will rarely have data for candidates that were *not* developed. Of course, the analogue is also true for forecasting new products, forecasting for assortment planning, or forecasting promotional sales: in each case, the dataset of newly introduced products or promotions that went live at the point of sale were decided upon based on some forecast that was high enough to clear an internal go/no go hurdle. This potential selection bias is rarely discussed in the literature.

4. Product-level demand forecasting in retail

Product-level demand forecasting in retail usually aims to generate forecasts for a large number of time series over a short forecasting horizon, in contrast to long-term forecasting for only one or a few time series at a more aggregate level. The ability to forecast the demand for each item in each store or DC as accurately as can be achieved is critical to the survival and growth of a retail chain, because many operational decisions, such as pricing, space allocation, listing/delisting, ordering and inventory management for an item, are related directly to its demand forecast. Forecast errors lead directly to poorer service and higher costs. Order decisions need to ensure that the inventory level is neither too high, in order to avoid high inventory costs (including spoilage and obsolescence), nor too low, so as to avoid stock-outs and lost sales. As a consequence, forecasting for inventory control should actually place more emphasis on quantile forecasts, predictive densities or prediction intervals, although most of the published research in this field still focuses on unbiased expectation forecasts or the minimization of MAPEs (see Section 5.1).

4.1. The hierarchical structure of product-level demand forecasting

In general, when given a decision-making question, we need to characterize the product demand forecasting question on three dimensions: the level in the product hierarchy, the position in the retail supply chain, and the time granularity (Fig. 5); these are sometimes labelled 'data cubes'.

4.1.1. Time dimension

Demand forecasts are needed at different time granularities for different managerial decisions. In general, the higher the level of the decision, from the operational to the strategic, the lower the forecasting time granularity. For example, we may need forecasts at a daily granularity for store replenishment and at a weekly level for DC replenishment, promotion planning, and (initial) allocation planning, while online fashion sales may rely on an initial estimate of total seasonal sales that is updated just once mid-season.

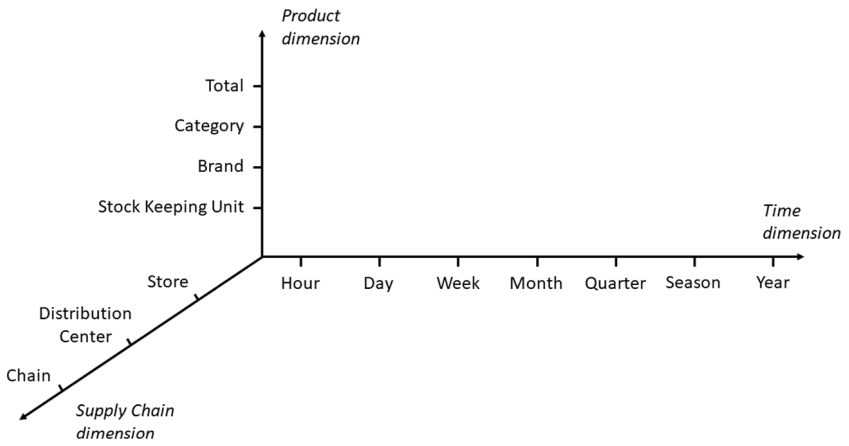


Fig. 5. Multidimensional hierarchies in retail forecasting. *Source:* Adapted from Syntetos, Babai, Boylan, Kolassa, and Nikolopoulos (2016).

4.1.2. Product dimension

Three levels of the product hierarchy are often used for planning by retailers: SKU level, brand level, and category level.

The SKU is the smallest unit for forecasting in retail, being the basic operational unit for planning daily stock replenishment and distribution. SKU-level forecasts are usually conducted within or across stores up to the chain as a whole and in daily/weekly time steps. The number of SKUs in a retail chain may well be huge. For example, a supermarket, drugstore or home improvement/do it yourself (DIY) retailer today has tens of thousands of items that need weekly or even daily forecasts. Walmart faces the problem of dealing with over one billion SKU × Store combinations (Seaman, 2018). In a fashion chain such as Zara, the number of in-store items by design, color and size can also be in the order of tens of thousands, although forecasting may be conducted at the "style" or design level, aggregating historical data across sizes and colors and disaggregating using size curves and proportions to arrive at the final SKU forecasts. Online assortments are typically far larger than those in brick-and-mortar stores, especially in the fashion, DIY and media (books, music, movies) businesses.

A brand in a product category often includes many variant SKUs with different package types, sizes, colors, or flavors. In addition to promotional planning, brand-level forecasts are also important where there are cross-brand effects, and promotions and ordering may be organized by brand.

However, for many retail decisions, the initial forecasts that are required are more aggregate, with a tactical promotional plan being developed across the chain that may well take inter-category constraints into account (although whether, in practice, forecasts have an active role in such a plan is an open question). A product category usually contains tens of brands and/or hundreds of SKUs that have certain attributes in common, e.g., canned soup, shampoo or nails. Categories may be segmented into subcategories, which may be either nested in or cut across brands. Category-level sales forecasting mainly focuses on weekly or monthly forecasts in a store, over a chain or across a market. Such forecasts are used chiefly for budget planning by so-called category managers, who make large-scale budgeting, planning and purchasing decisions, which again need to harmonize with the resources that are required to actually execute those decisions, e.g., shelf space, planograms or specialized infrastructure like available freezer space. The aggregation of SKU sales to weeks or months requires a consideration of trading days.

Category management and the assortment decision start with a category forecast, which Kök, Fisher, and Vaidyanathan (2015) suggested should be based on trend analysis supplemented by judgment. The assortment decision as to which brands (or SKUs) to exclude and which new products to add depends on the SKU-level demand forecasts: the effects on aggregate category sales of the product mix depend on the cross-elasticities of the within-category SKU-level demand forecasts, with a long (12-month) time horizon. Borin and Farris (1995) claimed that the associated shelf allocation is insensitive to SKU demand forecast errors.

In short, whatever the focus, SKU-level forecasts and their associated own and cross-price elasticities are needed to support both operational and tactical decisions.

4.1.3. Supply chain dimension

A typical retail supply chain consists of manufacturers, possibly wholesalers or other intermediaries, retailers' DCs, and stores in different formats. Retailers need forecasts of the demands faced by each level in the supply chain. SKU × store level forecasting is used for replenishment, sometimes driven by a top-down DC-level forecast that is allocated to stores on a historical basis. SKU × DC level forecasting drives the DC distribution/logistics

functions, while SKU × chain level forecasting is required for preordering. At the brand × chain level, forecasts are needed for supplier negotiations, as well as potentially for manufacturing decisions in vertically integrated retailers, such as many fashion chains.

A key question in retail supply chain forecasting is how to collaborate and integrate the data from different supply chain levels and actors, e.g., suppliers. From the retailer's perspective, such coordination, whilst costly, has the potential to improve availability and lower inventory. It may improve the retail forecasting accuracy or service levels (Wang & Xu, 2014), though some retailers doubt this, apparently only selling rather than sharing their data. Empirical models that analyze the relationship between POS data and the manufacturing forecast accuracy show that improvements are possible but not inevitable (Hartzel & Wood, 2017; Trapero, Kourentzes, & Fildes, 2012; Williams, Waller, Ahire, & Ferrier, 2014). Empirical evidence on successful retail implementation is limited, though Småros (2007) used case studies to identify some of the barriers and how they might be overcome, and Kaipia, Holmström, Småros, and Rajala (2017) further investigated the mechanisms that determine the value of introducing information sharing in collaborative sales and operations planning.

4.2. Forecasting within a product hierarchy

Given a specific retail decision-making question, we first need to determine the aggregation level for the output of the sales forecasting process. A common option is to choose a consistent level of aggregation for both data and analysis. For example, if one needs to produce demand forecasts at the SKU × weekly × DC level, it might seem "natural" to aggregate sales data to the SKU × weekly × DC level and forecast them at the same level. However, the forecasts can also be made using alternative forecasting processes within the data hierarchy: bottom-up, top-down, middle-out or hierarchical combination.

The choice of the appropriate level of aggregation depends on the underlying demand generation process. Existing research has found that the bottom-up approach is needed when there are large differences in structure between demand time series and the underlying drivers (Orcutt & Edwards, 2010; Zellner & Tobias, 2000; Zotteri & Kalchschmidt, 2007; Zotteri, Kalchschmidt, & Caniato, 2005), especially when the demand time series are driven by item-specific time-varied promotions. Foekens, Leeflang, and Wittink (1994) found that disaggregate models produced higher relative frequencies of statistically significant promotion effects, with magnitudes in the expected ranges. However, the top-down approach can generate more accurate forecasts in the case of many homogeneous demand series and small samples (Jin, Williams, Tokar, & Waller, 2015; Zotteri & Kalchschmidt, 2007; Zotteri et al., 2005). For instance, different brands of ice cream will have similar seasonalities, with a summer peak. While this may not be easy to detect for low-volume flavors, it can be estimated at a group level and applied at the product level (Syntetos et al., 2016). Song (2015) suggested that it is beneficial to model and forecast at the level of data at which stronger and more seasonal information can be collected.

Cluster analysis has been found to be useful for solving the trade-off, by improving the forecast performance (Boylan, Chen, Mohammadipour, & Syntetos, 2014). For example, when aggregating the product category level demand over stores, one can cluster stores based on whether they have similar demand patterns rather than according to their geographical proximity. A priori clustering based on store characteristics such as the size, range and location is common. Appropriately-implemented clustering can capture differences among stores (e.g., in terms of price sensitivity), as the clustering procedure groups stores with similar demand patterns (e.g., with similar reactions to price changes). In these terms, clustering is capable of resolving the trade-off between aggregate parameterization and heterogeneity, leading to more efficient solutions. So far, though, the majority of contributions on this issue have focused only on the use of aggregation for estimating seasonality factors (e.g. Chen & Boylan, 2007). These works have provided evidence that it can be helpful to aggregate correlated time series in order to estimate the seasonality better, since it can reduce the variability.

Hyndman, Ahmed, Athanasopoulos, and Shang (2011) proposed a method for reconciling forecasts of all series in a hierarchy optimally in order to ensure that they add up consistently over the hierarchy levels, with practical estimation methods also having been proposed (Wickramasuriya, Athanasopoulos, & Hyndman, 2018). Forecasts of all time series in the hierarchy are generated separately first, then these separate forecasts are combined using a linear transformation. A retail application by Oliveira and Ramos (2019) has examined the gains obtained by reconciling forecasts across the product hierarchy and found them to be substantial, particularly for longer lead times and higher levels of aggregation.

In general, hierarchical forecasting has received significant attention, but most researchers only consider the aggregation problem for general time series, without considering the specific characteristics of retail sales data, which are affected dramatically by many common factors, such as events, promotions and weather conditions. Research by Jin et al. (2015) suggested that regression-based methods, including many of the factors discussed above, produce substantially more accurate forecasts for store × SKU demand in promotion-intensive categories. At higher levels of aggregation, in both time and space, time series methods may well be adequate (Weller, Crone, & Fildes, 2016), though research for retail data remains to be done. However, there is not yet any straightforward answer as to how consistent demand forecasts can be generated feasibly on multiple interrelated large hierarchies over different dimensions.

4.3. Product-level retail sales data characteristics and drivers of demand

At the product level, many factors can affect the observed sales data and the characteristics of the potential demand. Some factors are part of the retailer's marketing mix plan (such as pricing and promotions, as well as "secondary" effects like interaction or cannibalization effects from listed, delisted or promoted substitute or complementary products), while other factors would be known in advance (such as sporting events, seasons and holidays). There are also some factors that may need to be predicted if they are to be included at all (such as competitive behavior, the local and national economy, and weather). There are many other unpredictable factors that can influence retail sales, such as anomalous events (for instance terrorist attacks or health scares) that will become random disturbances in the sales time series; their effects, along with those of weather and other events, may need to be removed carefully from the historical data prior to forecasting.

As a result of these diverse effects, product-level sales data are characterized by high levels of volatility and skewness; multiple seasonal cycles, especially when combined with 'special days' (e.g. bank holidays); often large volumes though with intermittence, with zero sales being observed frequently at the store level; and high dimensionality in any explanatory variable space. In addition, the data are also contaminated by stock-outs where the consumer is unable to purchase the product desired and instead may shift to another brand or pack size, or, in the extreme, leave to seek out a competitor retailer.

4.3.1. Stock-outs: demand vs. sales

Retail product-level demand forecasting usually depends on the SKU sales data captured by POS transactions. However, POS sales data form an imperfect observation of the true demand due to the demand censoring effect, where the actual demand exceeds the available inventory. Demand estimates that used only sales data would result in a negative bias in demand estimates of the focal product. At the same time, when facing a stock-out in the primary target product, customers may turn to purchase substitutes, which may increase the sales of substitute products and result in an overestimate of the demand for them. Academic researchers have long recognized the need to account for this censoring effect in inventory management. This body of literature has been centered primarily on methodologies for dealing with the imperfect demand observations. These methods can be classified into two categories: nonparametric methods (e.g., Kaplan & Meier, 1958) and parametric models using hazard rate techniques (e.g., Agrawal & Smith, 1996; Nahmias, 1994; Wecker, 1978). For more detail, see Tan and Karabati (2004), who provided a review on the estimation of demand distributions with unobservable lost sales for inventory control. Sachs and Minner (2014) compared parametric and nonparametric modeling approaches for estimating different censoring levels using data from a large European retail chain. They found that the nonparametric approaches are better than the parametric ones, with lower inventories for the same service levels. Most of the methods are based on stock-out events data, but Jain, Rudi, and Wang (2014) found that stock-out timing could improve the estimation accuracy further relative to methods based on stock-out events. In the marketing and assortment management literature, researchers have focused on consumers' substitution behaviors when their target product is facing a stock-out (e.g., Conlon & Mortimer, 2013; Kök & Fisher, 2007; Vulcano, van Ryzin, & Ratliff, 2012).

Conversely, there is some evidence that, at least for some categories, demand depends on inventory, with higher inventory levels driving higher sales; this has been called a "billboard effect" (Koschat, 2008; Ton & Raman, 2010). Anecdotally, we have encountered retailers who know this putative effect as "product pressure". However, no studies to date appear to have used inventories as a driver for improving forecasts.

In general, the forecasting models proposed in this area are explanatory rather than predictive, and they often require more information than is readily available, such as periodic stock auditing, customer numbers and assortment information. In addition, any forecasting algorithm that leverages system inventory information needs to deal with the fact that system inventories are notoriously inaccurate (so-called "inventory record inaccuracy" or IRI; DeHoratius & Raman, 2008). As a consequence, the models that have been published thus far are not suitable for forecasting applications. The limited research that has been reported in the forecasting literature may be due in part to the lack of real demand observations, meaning that the forecasting accuracy is hard to measure. On the other hand, storing the observed changes in the shelf inventory for every product may be very costly for retailers, and may not be adequate for identifying every single stock-out instance. Technological solutions such as RFID may become more common (Bottani, Bertolini, Rizzi, & Romagnoli, 2017). In a study of the online demand at Amazon, adopting a simple model for replacing stock-out periods led to major changes in the quantile forecasts, which had a direct effect on ordering (Seeger, Salinas, & Flunkert, 2016). In a recent case study of an e-grocery retailer, the customers' online ordering process could monitor customer preferences before stock-out information became available to the buyer, thus yielding uncensored demand data (Ulrich, Jahnke, Langrock, Pesch, & Senge, 2019).

4.3.2. Intermittence

Intermittence – whether driven by low demand or by stock-outs – is another common characteristic of store POS sales data, especially at the SKU $\times$ daily level. Fig. 6 depicts an SBC (Syntetos, Boylan, & Croston, 2005) categorization (see also Kostenko & Hyndman, 2006) over the daily sales of 1373 household cleaning items from a UK retailer, cross-classified by the coefficient of variation in demand and the mean period between non-zero sales. A total of 861 items exhibit strong intermittent characteristics.

Techniques designed specifically for intermittent demand include Croston's method (Croston, 1972), the bootstrap of Willemain, Smart, and Schwarz (2004), the

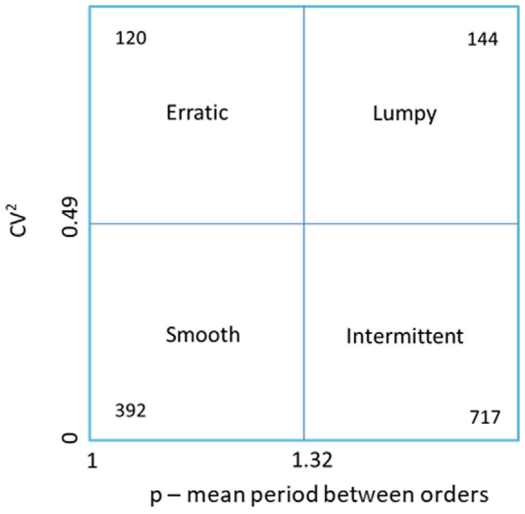


Fig. 6. SBC categorization of 1373 household cleaning items by the coefficient of variation and the mean period between non-zero daily demands.
Source: UK supermarket data.

Syntetos–Boylan approximation (SBA) method (Syntetos & Boylan, 2005), and the TSB method (Teunter, Syntetos, & Babai, 2011). However, most of these models have been tested on demand/sales time series data from industries other than retail (e.g., service/spare parts, or high-priced capital goods in electronics, automotive, aerospace and high tech), with the exception of the study by Kolassa (2016), who found that Croston's method point forecasts with a negative binomial distributional assumption yielded competitive density forecasts. Croston's method is intuitively appealing, has some theoretical justification (Hyndman, Koehler, Ord, & Snyder, 2008, p. 282) and is used commonly in practice, at least as a benchmark. However, Shenstone and Hyndman (2005) and Snyder (2002) point out certain inconsistencies; i.e., models that might be considered as underlying Croston's methods are inconsistent with the properties of intermittent demand data. Nevertheless, Shenstone and Hyndman (2005) noted that Croston's point forecasts and prediction intervals may still be useful.

Product-level demand in retail is also disturbed by a number of exogenous factors, such as promotions, special events, seasonalities, weather, etc. (as will be discussed in what follows), all of which factors make intermittent demand models difficult to apply to POS sales data. One possibility is to model these influences on intermittent demands via Poisson or negative binomial regression, although Kolassa (2016) found that the best models included only day of the week patterns. One alternative approach is the use of time series aggregation in the multiple aggregation prediction algorithm (MAPA; Kourentzes, Petropoulos, & Trapero, 2014) to overcome intermittence. Kourentzes and Petropoulos (2016) applied the method to a manufacturer's cider sales, generalizing it to include multiple retailers' promotions as explanatory variables, and found that it performed better than benchmarks such as exponential smoothing with explanatory variables. MAPA(X) has not yet been applied to SKU × store level retail time series. Li and Lim (2018) proposed a greedy aggregation–decomposition method, which utilizes both aggregate and disaggregate forecasts to improve the forecasting accuracy of SKU-store-day demand.

4.3.3. Seasonality

Retail product sales data have strong seasonality and usually contain multiple seasonal cycles of different lengths. For example, the beer sales data shown in Fig. 7 exhibit both weekly and annual cycles. Sales tend to be high during the weekend and low on weekdays, high in summer and low in winter, and high around Christmas. Some sales data may also possess biweekly or monthly (paycheck effects) or even quarterly seasonalities, depending on the nature of the business and business locations. For this reason, models that are being used for forecasting must be able to handle multiple seasonal patterns. Huang, Fildes, and Soopramanien (2019) demonstrated this point, using models with sufficient flexibility but parsimonious complexity to capture the seasonality of weekly retail data: trigonometric functions proved sufficient. Alternatively, one can include Boolean dummies for weeks in the year (e.g., Huang, Fildes, & Soopramanien, 2014), although this typically leads to a 'degrees of freedom' problem and ignores the regularity from period to period.

4.3.4. Calendar events

Retail sales data are affected strongly by some calendar events. These events may include holidays (Fig. 7 shows a significant lift at Christmas, i.e., in week 51), festivals, and special activities (e.g., important sport matches or local activities). For example, Divakar, Ratchford, and Shankar (2005) found that the demand for beverages increased substantially during holidays, while other product groups were affected negatively. In addition, SKU × Store consumption may change due to temporary changes in local demographics. Certain holidays recur at regular intervals and can thus be modelled as seasonality, e.g., Christmas or the Fourth of July in the US. Other holidays move around more or less widely in the (Western style) calendar and therefore cannot be captured as seasonality, such as Easter, Labor Day in the US, or various religious holidays of which the date is determined based on non-Western calendars, such as the Jewish or Muslim lunar calendars. Such holidays can be modeled using dummies (Cooper, Baron, Levy, Swisher, & Gogos, 1999).

4.3.5. Weather

The demand for some retail products is also affected strongly by temperature and other weather conditions. For example, there is usually strong support for the idea that the sales of soft drinks are higher when the weather is hot, although much of the literature has assessed only the in-sample fits or statistical significance of weather predictors (Dubé, 2004; Murray, Di Muro, Finn, & Popkowski Leszczyc, 2010). Nikolopoulos and Fildes (2013) showed how a brewing company's simple exponential

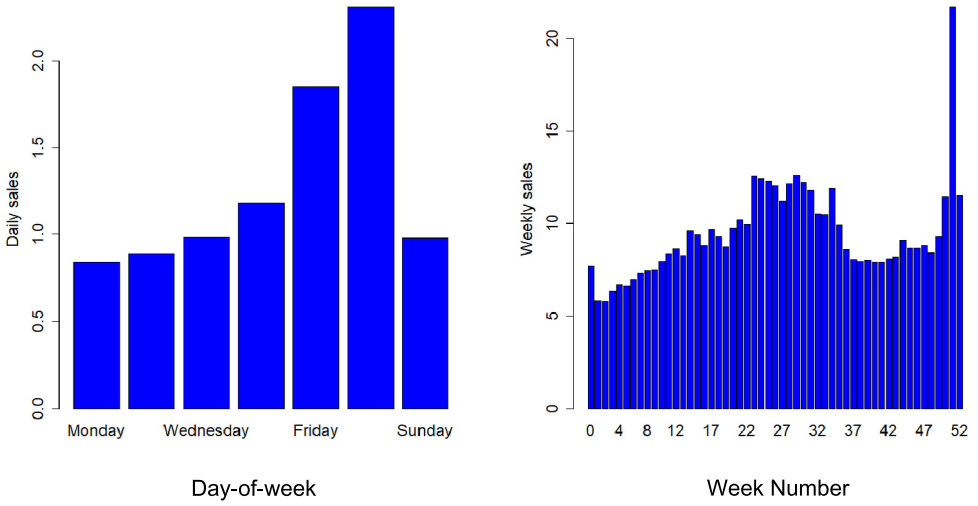


Fig. 7. Beer daily and weekly sales: UK supermarket data.

smoothing method for in-house retail SKU sales could be adjusted (outside the base statistical forecasts) to take into account temperature effects. In an examination of weather effects on sales in an online fashion retailer, Steinker, Hoberg, and Thonemann (2017) found that including weather drivers achieved substantial accuracy improvements of around 10%, leading to cost savings of 11.6% through better workforce planning.

Weather effects may well be nonlinear. For instance, sales of soft drinks as a function of temperature will usually be flat for low to medium temperatures, then increase with hotter weather; however, the increase may taper off again with extreme heat, when people switch from sugary soft drinks to straight water. In principle, such effects could be modeled using spline transformations of temperature.

One challenge of using weather data to improve retail sales forecasts is that there is a plethora of weather variables available, though the available evidence suggests that temperature is the most effective. There is the additional possibility of differential effects as the year progresses, where "the first barbecue weather of the season" may have a higher impact than similar weather later in the year.

Of course, another hurdle is that weather variables themselves need to be forecasted, unlike intervention variables like prices or promotions that the retailer sets themselves, or calendar events of which the dates are known with certainty. This means that weather data can be used meaningfully only for either producing short-range sales forecasts, since weather forecasts are better than chance only for short horizons, or cleaning *past* data of the historical impacts of, say, heat waves or hurricanes. This implies that forecasting exercises that use the actual weather in *ex post* forecasting appraisals overstate the forecast's certainty, since they do not include the uncertainty that is inherent in the weather forecast. In principle, this uncertainty can be surmounted in analyses; however, it has been our experience that it is much more expensive to obtain historical weather *forecasts* from data providers than historical *actual weather data*. In addition, one needs to ensure the correct vintage of forecasts: when calculating two-day-ahead sales forecasts, we need two-day-ahead weather forecasts, for three-day-ahead sales forecasts, we need three-day-ahead weather forecasts, and so on.

4.3.6. Marketing mix and promotions

The regular price and relative price discount are important variables that should be included in any forecasting model. When models focus on short-term forecasts, the promotional price is important. Apart from price effects, the effects of feature advertisement have been studied extensively in the marketing literature. Feature advertisements can be divided into in-store advertisements and other advertisements like using a newspaper or a store flyer to increase the store traffic (Gijsbrechts, Campo, & Goossens, 2003). In-store advertisement focuses on attracting customers to the promoted articles in several ways, with the most commonly used methods making use of ads and displays (Cooper et al., 1999). The effects of displays have also been investigated intensively in the marketing literature. A general conclusion is that sales can increase several-fold in the presence of displays (Ailawadi, Harlam, César, & Trounce, 2006).

Fig. 8, which is a result of a data exploration of the IRI retail data (Bronnenberg, Kruger, & Mela, 2008; note that "IRI" here refers to the eponymous company), shows that the different marketing instruments have different magnitudes of lift effects on different product categories. Categories with long shelf lives lend themselves to stockpiling or "pantry loading", resulting in stronger promotional uplifts and post-promotional dips than those seen for products with shorter shelf lives, like the fresh and ultra-fresh categories.

In addition, different promotional types (e.g. buy-one-get-one-free (BOGOF) versus 50% price reduction) have effects that are not captured just by the unit price or small

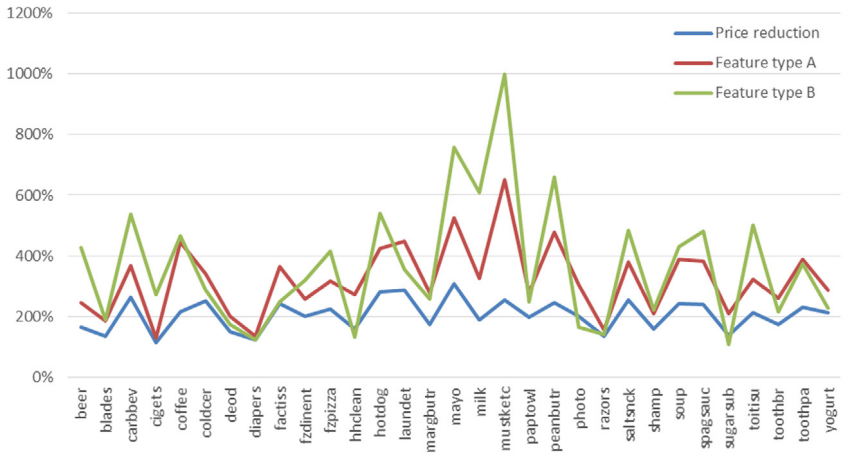


Fig. 8. Promotional lift effects in the IRI dataset among various categories. The vertical axis is the ratio of the sales during the periods with promotions to those in the periods without promotions.
Source: IRI data set.

numbers of different feature types, and marketing instruments may also be used in combination, with pronounced interaction effects. These effects do not appear to have been addressed in forecasting research, although some retail forecasting software products do model such interactions and nonlinearities. Fig. 9 shows a typical situation with different tactics interacting.

The quest for novelty on the part of retailer marketers also ensures that it will be common for products to need to be forecasted with a marketing mix that has not been observed previously for that particular product. This necessitates a kind of "new promotion" forecasting that is analogous to a certain extent to new store and new product forecasting. This does not seem to have been addressed in the literature to date, and would be an interesting avenue for future research.

The majority of the promotional response stems from either brand switching or accelerating or "pulling forward" future purchases in the store (Bucklin & Siddarth, 1998; Chiang, 1991; Chintagunta, 1993; Gupta, 1988). That means that a promotion on one item may affect either the sales of another item or its own later sales, so-called "cannibalization". A large body of research supports the view that brands within a product category are substitutes for one another (Kumar & Leone, 1988; Moriarty, 1985; Mulhern & Leone, 1991; Walters, 1988, 1991), and the incorporation of cannibalization effects can improve forecasts substantially (Srinivasan, Ramakrishnan, & Grasman, 2005), at least at the aggregate level. However, it is not clear whether the cannibalization signal at the SKU $\times$ store level is strong enough to improve the forecasts at this granular level and to actually improve the stock position on the shelf. In addition, the modeling of cannibalization requires a significant additional effort for identifying drivers and victims, although product hierarchies may help here, and therefore a retailer would be likely to restrict the modeling of cannibalization to important categories. Finally, much the same discussion as for cannibalization applies to interaction effects from complementary products. For instance, it may be hypothesized that a promotion on steaks will increase the sales of steak sauces.

The analysis of these two types of interaction – cannibalization and complementarity – differs in two key aspects. First, as was noted above, product categories typically group similar products that are likely to be substitutes, meaning that the product category can be used to identify potentially-interacting pairs or groups of products. Conversely, the product hierarchy typically does not group complementary articles together, so it is not useful for identifying pairs or groups of products that may exhibit complementarity. Second, basket analysis (i.e., the analysis of transaction log data with a view to examining which products were bought by the same shopper at the same time) can be useful in detecting complements, using affinity analysis; however, it is harder to use basket analysis to detect substitutes.

However, the cross-category price elasticities appear to be small, limiting the scope for improving forecasts using complementarity (Russell & Petersen, 2000). Ma, Fildes, and Huang (2016) showed that forecast accuracy improvements of 12% (as measured by the mean absolute error) can be captured by the inclusion of competitive effects, while an additional 0.6% can be obtained with cross-category effects.

The large variety of promotion types and complex promotion-interaction effects make product sales difficult to forecast in formal model-based approaches. Methods exist for incorporating the large number of promotion-related variables (e.g. Ma et al., 2016) into an operational forecasting model, as is necessary for retail promotion optimization, and software solutions exist that implement such models, e.g. the SAP Customer Activity Repository solution for retail. However, there is little evidence of any widespread use of such models. Instead, many retailers use a simple statistical model supplemented by

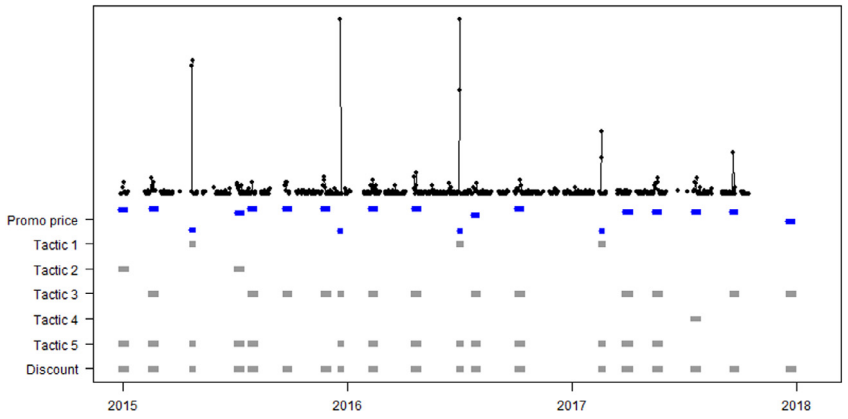


Fig. 9. Daily sales of one SKU in one store (black time series plot at the top, *y*-axis annotation suppressed for confidentiality reasons), promotional prices (blue in the middle, with nonpromotional prices left out for clarity), and indicators of five different tactics and a discount (six Boolean indicators at bottom). Note the use of many different tactics and extremely different promotional uplifts. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

judgmental adjustments from the demand planning team. An important research question is what (if any) benefits accrue from the increasingly complex alternative methods and the ways in which demand planners interpret and interact with such models.

4.3.7. Online product reviews and social media

In recent years, online product reviews have been found to be an important source of market research information for online retailers (Floyd, Ling, Alhogail, Cho, & Freling, 2014), and we speculate that this applies to all retailers for which service is an important component. Since such reviews are a voluntary expression of consumers' experiences and beliefs about the quality of products and services, consumers often rely on online product reviews when making their own purchasing decisions (Chen & Xie, 2008; Zhao, Yang, Narayan, & Zhao, 2013). Researchers have found that there is a strong relationship between online word-of-mouth and product sales, although the impact of word-of-mouth varies with the product category (Archak, Ghose, & Ipeirotis, 2010; Chen, Wang, & Xie, 2011; Chevalier & Mayzlin, 2006; Hu, Koh, & Reddy, 2014; Zhu & Zhang, 2010). Thus, the incorporation of online product reviews, using tools such as text mining and sentiment analysis, may allow online retailers to add a new layer to their existing predictive models and boost predictive accuracy.

Product reviews are mainly textual data, which cannot be used directly in a sales forecasting model, although numerical summaries are common. The basic idea of using product reviews as explanatory variables is to identify the extent of the product review's polarity, e.g., strongly positive, strongly negative, or neutral. Ratings are a simple form of product review, and retailers often adopt a five-star rating mechanism, e.g. Amazon's product rating system. This method is simple and fast. Another way of detecting the polarity of a review is to examine the textual content of the review (Ku, Lo, & Chen, 2007). In this case, a keyword dictionary needs to be established before the polarity of reviews can be analyzed using text mining and sentiment analysis techniques (Chern, Wei, Shen, & Fan, 2015; Lau, Zhang, & Xu, 2018). An alternative way of using product reviews is to represent text in a bag-of-words model, dealing with the resulting large number of predictors using, say, random projection (Schneider & Gupta, 2016). However, once measured, online reviews do not generate a uniquely important driver variable by interacting with any promotions; despite this limitation, both online reviews and their interactions with promotions have proved important for forecasting as demonstrated in an application to Amazon online sales of electronic products (Chong, Li, Ngai, Ch'Ng, & Lee, 2016). However, it should be noted that fake reviews and so-called "sock puppetry" are a concern (Zhuang, Cui, & Peng, 2018), and the automatic detection of such fake reviews is an active field of study (e.g., Kumar, Venugopal, Qiu, & Kumar, 2018). Similarly, there are concerns that product reviews may be biased through herding behaviors, affecting the resultant summary measures (Schaer, Kourentzes, & Fildes, 2019).

Similar challenges arise if we try to improve retail sales forecasts using social media data, such as Facebook, Twitter, Weibo, blogs or similar user-generated content. Here, a social media post first needs to be matched to the corresponding product, unlike online product reviews on a product's page, where it is clear which product a review belongs to. Once this step has been taken, similar text mining methods can be brought to bear on this topic as in the case of online product reviews. For instance, Cui, Gallino, Moreno, and Zhang (2018) found that the inclusion of social media information improved daily sales forecasts for an online men's fashion retailer, and that this improvement was robust across a number of different causal forecasting methods, though the results are based only on three closely-related aggregated time series. Care must be taken to distinguish forecasts that use user-generated social media data from forecasts that use social

media data which the retailer (or the manufacturer) created themselves as part of conducting marketing activities on social media (Kumar, Choi, & Greene, 2016).

Given the ephemerality of social media and the difficulty of forecasting using customer reviews or social media posts themselves, in contrast to a retailer's own marketing and pricing activities, these variables probably offer only limited possibilities for improving short-term forecasts, not longer term ones. Kalampokis, Tambouris, and Tarabanis (2013) and Schoen et al. (2013) reviewed the forecasting of product sales (and other variables of interest) using social media data; they drew similar conclusions and commented on the inadequate evaluations that have been carried out. A recent and updated evaluation is provided by Schaer et al. (2019), who reached the same conclusions.

4.4. Data pooling

Once an appropriate aggregation level has been selected, another modeling issue is to decide the appropriate extent to which to pool the models and data across stores and SKUs. Pooling sales data across many items together clearly enhances the data that are available for modeling and has the potential to capture common patterns across time series. Such an approach may be expected to lead to a higher forecasting accuracy, with fewer parameters to be estimated, allowing a more rapid adaptation to any structural changes in time series, and with robustness in the presence of outlier observations (e.g., Dekker, van Donselaar, & Ouwehand, 2004; Duncan, Gorr, & Szczypula, 2001). In addition, pooling enlarges the observed ranges for the explanatory variables. Marketing mix elasticities and seasonality patterns are usually assumed to be homogeneous in a pool of stores, but the baseline sales are allowed to be heterogeneous for different stores (Ainscough & Aronson, 1999; Baltas, 2005; van Donselaar, Peters, de Jong, & Broekmeulen, 2016). However, the decision as to which variables should be assumed to be homogeneous (or heterogeneous) is still a matter of judgment. The downside of inappropriately assuming homogeneity is that the forecast equations are mis-specified, with a resulting bias.

In the famous PromoCast model, Cooper et al. (1999) used a 67-variable cross-sectional pooled regression analysis of SKU–store sales under a variety of promotional conditions with store- and chain-specific historical performance information. Andrews, Currim, Leeflang, and Lim (2008) found that accommodating store-level heterogeneity did not improve the accuracy of marketing mix elasticities relative to the homogeneous model, and the improvements in the fit and forecasting accuracy were also modest. Gür Ali, Sayin, van Woensel, and Fransoo (2009) compared the accuracies of 30 SKU sales prediction methods that differed in their data richness, technique complexity and model scope using a multi-store, multi-SKU European grocery sales database. They found that pooling observations across stores and subcategories provided better predictions than pooling across either only stores or only subcategories.

In contrast to the findings of Andrews et al. (2008) and Gür Ali et al. (2009) that store-homogeneous models provided better forecasts, Lang, Steiner, Weber, and Wechselberger (2015) found that allowing for heterogeneity in addition to functional flexibility (P-splines instead of linear) could improve the predictive performance of a store sales model considerably, but that incorporating heterogeneity alone only moderately improved or sometimes even decreased the predictive validity.

Another benefit of data pooling is that it can be used to forecast the demand for new SKU–store combinations that are not present in the training data. This can then be used to make allocation decisions for both existing and new stores. For example, Gür Ali (2013) proposed a "Driver Moderator" method which generated short-term forecasts for both existing and new SKUs by pooling information across SKUs and stores. Similarly, Ferreira, Lee, and Simchi-Levi (2015) presented a pricing DSS for an online retailer, Rue La La, which offered extremely limited-time discounts (“flash sales”) on designer apparel and accessories. They used the features of the historical data to build a regression model for each product department, pooling data starting from a SKU such as women's athletic shoes up to the top of the hierarchy, the corresponding department, such as footwear. The model then predicted the demands for styles that were being sold for the first time; these depended on the price and the level of discount, as well as the SKU characteristics. While the forecast evaluation was cross-sectional, the successful revenue optimization experiment supported the effectiveness of the demand model.

In addition to data pooling, studies have also found that forecasts could be improved further by mining the residuals from many SKUs pooled across subcategories and stores. Based on the PromoCast model proposed by Cooper et al. (1999), Cooper and Giuffrida (2000) used data mining techniques on the residuals to extract information from many-valued nominal variables, such as the manufacturer or merchandise category. The output of their data mining algorithm was a set of rules that specified what adjustments should be made to the forecast produced by the homogeneous market-response model. This combination meant that a more complete array of information could be used for the development of tactical planning forecasts. Trusov, Bodapati, and Cooper (2006) further improved the accuracy of the forecasts and the interpretability of the recommendation system for promotional forecasts. Gür Ali and Pinar (2016) proposed a two-stage information-sharing method. First, segment-specific panel regressions with seasonality and marketing variables pooled the data, then the residuals were extrapolated non-parametrically using features that were constructed from the last twelve months of observations from the focal and related category–store time series. The forecast combined the extrapolated residuals with the forecasts from the first stage, showing out-of-sample accuracy (measured by MAE, MAPE and MdAPE) improvements of 15% to 30% over horizons of 1 to 12 months compared with that of the one-stage model. Exponential smoothing provided the benchmark, where the gains again proved substantial (between 25% and 40%).

When there is not sufficient item-level data available to develop an item-specific model, the research summarized above has shown that pooling is a good way of improving forecasts. However, if there is enough historical data available for a single SKU, the following questions arise: should we use an individual model for that SKU or a pooled model that considers all other SKUs? And how much historical data is enough? Though Trapero, Kourentzes, and Fildes (2014) found that pooled models were better for SKUs that had fewer than five observations on promotions, this finding cannot necessarily be generalized, as many types of promotions are used in practice (cf. Fig. 9), and the results may also depend on the models employed. Thus, we call for more systematic research to answer these questions.

4.5. Dealing with high dimensionality in the explanatory variable space

Any complete specification of the product- (SKU-)level determinants of store sales has high dimensionality in the explanatory variable space due to cross-item promotional interactions, which pose a big challenge in product demand forecasting. The high dimensionality stems mainly from competing products within the same category (Ma et al., 2016). In such situations, the model may be over-fitted easily, or perhaps cannot even be estimated.

One simple solution is to select the most influential subset of items from the same product category as the focal product. For example, the forecasting model named CHAN4CAST, developed by Divakar et al. (2005), considered only the main competitor's promotional variables; i.e., considering only Pepsi's promotions when forecasting the sales of Coca-Cola. Similarly, Lang et al. (2015) selected the lowest price of a competing national (premium) brand in-store as representing the competition. Ma et al. (2016) used the promotional information from the top five sales products in the same category as the focal SKU, resulting in a 6.7% improvement in forecast accuracy (measured by MAE) compared to models that used only information on the focal variable.

Another way of overcoming the dimensionality problem is to build predictive models based on summaries of the cross-promotional effects. The basic idea of this method is to build indexes which could summarize the cross-promotional information. For example, van Donselaar et al. (2016) simply used the number of SKUs in a promotion as a summary of the promotional intensity in the category. Another straightforward way of building promotional indexes is to construct a weighted averaging of the promotion values (discount, display and feature) across SKUs in the category (e.g., Natter, Reutterer, Mild, & Taudes, 2007). Voleti, Kopalle, and Ghosh (2015) proposed a more elaborate approach by incorporating branding hierarchy effects and inter-product similarity simultaneously.

The third way is to summarize the promotional information by extracting a few diffusion "factors" by means of principal component analysis (PCA; Stock & Watson, 2002). One criticism of factor-augmented regressions is that the factors are estimated without taking into account the dependent variable. Thus, when only a few factors are retained to represent the variations of the whole explanatory variable space, they might have only limited predictive power for the dependent variable, whereas the discarded factors might be more useful.

Another solution is based on variable selection, especially using penalized likelihood methods, where influential promotion variables are selected automatically via continuous shrinkage (Gür Ali, 2013; Gür Ali & Yaman, 2013; Huang et al., 2014; Ma et al., 2016). Traditional best subset selection procedures are usually not feasible for high-dimensional data analysis because of the high computational cost. Penalized likelihood methods for coping with high dimensionality have been developed successfully over the last decades (Friedman, 2012; Tibshirani, 2011). A number of recent studies on product demand forecasting have been based on this method. For example, Gür Ali (2013) proposed a "driver moderator" method which uses basic SKU–store information and historical sales and promotion data to generate many features, and an L1-norm regularized regression to simultaneously select a few relevant features and estimate their parameters. Similarly, Huang et al. (2014) also identified the most relevant explanatory variables using L1-norm regularized methods. Ma et al. (2016) proposed a four-step methodological framework which consists of identifying potentially influential categories, building the explanatory variable space, conducting variable selection and model estimation by means of a multistage LASSO (least absolute shrinkage and selection operator) regression, and generating forecasts. The success of this method for dealing with high dimensionality was demonstrated by substantial improvements in forecasting accuracy compared to alternative methods of simplifying the variable space. The multi-stage procedure overcomes the known limitation of LASSO when dealing with highly correlated explanatory variables.

While the issue of dimensionality reduction of the explanatory variable space may seem arcane, simple methods such as stepwise regression do not work, and the effects on both forecast accuracy and overall profitability (through a retail price-promotion optimization) are substantial (Kunz & Crone, 2015). Practical research is needed for dealing with the typical messy data that are encountered in retail operations where time series histories are short, multiple promotion types are relevant and product assortments change. There are potentially major benefits to be gained from estimating the loss from just using simpler methods.

4.6. Quantile and density forecasts for replenishment

The main use for SKU-level forecasts is replenishment. As a consequence, quantile forecasts potentially are even more important than expectation forecasts. This is complicated by the fact that standard inventory theory presupposes normally-distributed and possibly even stationary demand, both of which are unrealistic when the demand is intermittent and promotion-driven.

Taylor (2007) developed an exponentially-weighted quantile regression method that generates interval forecasts from quantile predictions. However, these methods do not take into account external factors such as price changes and promotions. Beutel and Minner (2012) propose two different approaches that include predictor information: on the one hand, a regression forecast that would feed into an inventory control system, and, on the other, a linear programming model that sets target stocks directly without explicit forecasting.

An alternative would be to forecast full predictive densities, from which quantiles can be extracted easily. Kolassa (2016) did so, using a variety of parametric and nonparametric methods to model weekly and yearly seasonality and price effects. Sillanpää and Liesiö (2018) predicted the full distribution of intermittent retail consumer demands in order to forecast store orders transmitted to the DC better. Their generalized linear models only included the day of the week, but could have accommodated promotions and other drivers as predictors as well. Machine learning methods do not deliver predictive densities automatically (in contrast to parametric statistical methods); solutions to this are discussed in Section 5.5.

5. An evaluation of product-level demand forecasting methods

Over the past several decades, a considerable amount of effort has been devoted to the development and improvement of demand forecasting methods in retail. Beyond well-established univariate extrapolative methods such as exponential smoothing, linear regressions (and variants) that include various driver variables are better established than more complex methods. Such linear models have the important practical advantage of being easy to interpret and implement. On the other hand, more complex nonlinear models should be considered if linear models fail to perform well (Chu & Zhang, 2003). Indeed, they are embedded in some commercial software already (Fildes, Schaer, & Svetunkov, 2018). We review these different classes of methods here, after first considering specific issues arising with their evaluation when applied to product demand.

5.1. Forecast evaluation at the product level

As is the case in all forecasting subdisciplines, forecasts in retail need to be evaluated. The standard forecast accuracy metrics are commonly employed; however, the specific challenges in retail, particularly at the product level, imply that some metrics may be misleading or even unusable. This depends mainly on the level of aggregation.

One commonly-employed accuracy metric, as we see in our retailing vignettes discussed in Section 7 and Appendix B, is the mean absolute percentage error (MAPE), due mainly to its ease of interpretation and comparability between series on different scales. This is especially important for practitioners who need to explain a forecast and its accuracy to non-specialists in forecasting. However, the MAPE can only be used on sufficiently highly aggregated time series, as it is undefined if there are zero realizations. A related scale-free measure is the weighted MAPE (wMAPE), which can also be expressed as the ratio of the mean absolute error (MAE) to the mean of the actual observations (Kolassa & Schütz, 2007).

Non-scale-free accuracy measures include the mean absolute error (MAE) or mean absolute deviation (MAD), and the mean squared error (MSE) or root MSE (RMSE). These usually are not suitable for use in retail forecasting, where we typically have multiple time series on very different scales. One approach to making the MAE or (R)MSE scale-free is to scale it by an appropriate factor, such as the mean of the actuals in the evaluation period, or the overall mean. An alternative is to scale them by the corresponding error of a benchmark forecasting method, leading to *relative* errors with respect to this particular benchmark, e.g. the relative MAE (RelMAE) and relative MSE (RelMSE) or the MASE (Hyndman & Koehler, 2006). Measuring the forecast improvement of one method over another is particularly important when considering the "forecast value added" when managerial judgment is used to adjust the statistical forecast (see Davydenko & Fildes, 2013, and the discussion in Section 5.3).

However, one problem with almost all of these accuracy measures has been very much underappreciated. Specifically, since forecasting inherently deals with uncertainty, let us "take a step back" and consider the predictive density that is considered, explicitly or implicitly, underlying every forecasting method. Given this density, each point forecast accuracy measure is minimized in expectation by a specific function of the predictive density (Gneiting, 2011a; Kolassa, 2019). Thus, if the predictive density is asymmetric, minimizing the MAE may lead to biased forecasts. Even for only somewhat intermittent demand, the MAE has been found frequently to be minimized by an – unusable – flat zero forecast (Kolassa, 2016; Morlidge, 2015). This implicit bias also afflicts derivatives of the MAE, like the relative MAE (RelMAE) and the MASE. The expected MAPE in turn is minimized by the '(-1)-median' of the predictive density (Gneiting, 2011a, p. 748), which is typically lower than the expectation even for symmetric densities, especially in the case of data with high coefficients of variation (Kolassa, 2017).

Thus, we should keep in mind that common point forecast accuracy measures may appear to be easy to interpret, but that actually this may be misleading, especially for data of a very fine granularity. Forecasters are well advised to report bias measures along with MAPEs or similar KPIs, or to use measures directly that are minimized in expectation by an unbiased forecast, like the MSE or RelMSE.

A recent proposal for addressing these shortcomings of classical point forecast accuracy KPIs, especially for intermittent demands, is to use rate-based errors (Kourentzes, 2014), which assess whether forecasts for an intermittent series are correct cumulatively over increasing time horizons. As yet, though, there seems to have been little adoption of these measures.

In addition, not only are expectation point forecasts important, but also high quantile forecasts, especially when forecasts are used for replenishment and for determining safety stock amounts. Gneiting (2011b) discusses

appropriate accuracy measures, which, to the best of our knowledge, have not been taken up in the retail forecasting literature to date. Note also that replenishment cycles may cover multiple forecasting time buckets, meaning that the calculation of safety amounts requires quantiles of sum forecasts, or of convolutions of predictive densities. Kolassa (2016) argues that, as a consequence, forecasters should aim to forecast and evaluate the entire predictive density, not least because of the potential use of such densities in replenishment, as was discussed in Section 4.1.

5.2. Univariate forecasting methods

The basic product-level demand forecasting methods are univariate forecasting models that use only the past sales history. The techniques used in retail product demand forecasting range from traditional time series techniques, such as simple moving averages, the exponential smoothing family or ARIMA, and Fourier analysis (e.g., Fumi, Pepe, Scarabotti, & Schiraldi, 2013), to state space models (e.g., Ramos, Santos, & Rebelo, 2015, who compared these models to ARIMA). Researchers such as Gür Ali et al. (2009) and Ramos and Fildes (2017) have found that simple time series techniques performed well for periods without focal product promotions. However, methods with promotional drivers improved the accuracy substantially for periods with promotions. Thus, univariate forecasting methods should be adopted only for higher aggregation demand forecasting (e.g., product category demand at a chain) or for products with a low promotional intensity or price elasticity of demand.

5.3. Base-times-lift and judgmental adjustments

In practice, many retailers use a base-times-lift approach to forecasting product demand (Cooper et al., 1999). The approach is usually a two-step procedure which begins by generating a baseline forecast from a simple time series forecasting method where, in principle, the data should have been cleansed of the demand drivers such as past promotions, weather etc. This baseline forecast is then adjusted for any upcoming events, most often promotions (Fildes & Goodwin, 2007). The adjustments for events are often estimated based on the lift effect of the most recent similar event (e.g. price reduction and/or promotion) and the judgements made by retail managers (Cooper et al., 1999). We conjecture that this is due in part to the installed base of commercial software, rather than to any appraisal of the effectiveness of more complex algorithms. A more comprehensive approach is to estimate (perhaps analytically) the promotional profile and the expected impact of any special events such as the start of the school holidays, and add these adjustments to the baseline.

The use of analogous past promotions that are regarded as similar to the forthcoming promotion is a natural basis for a forecast (Fildes & Goodwin, 2007; Lee, Goodwin, Fildes, Nikolopoulos, & Lawrence, 2007). McIntyre, Achabal, and Miller (1993) proposed a case-based reasoning system for facilitating promotional adjustments. The system selected the historical analogs that were most similar to the planned promotion, then adjusted the sales of each analog to account for any differences between the analog and the planned promotion. The forecasts were derived from the multiple analogs so as to arrive at a single sales projection. Lee et al. (2007) conducted an experiment which showed that a forecasting support system (FSS) could be designed for providing users with guidance on appropriate similarity judgments based on evidence from past promotions; the resulting adjustments led to more accurate forecasts of the effects of sales promotions. However, Fildes, Goodwin, and Önkal (2019) showed in experiments that such base lift adjustments fail to take into account the full history of similar product promotions, resulting in an increase in forecast errors.

Judgmental adjustments are common in practice, but expensive for SKU-level demand forecasting where the number of adjustments is large and their size and direction are prone to systematic biases and inefficiencies (Fildes & Goodwin, 2007; Fildes, Goodwin, Lawrence, & Nikolopoulos, 2009). There is a substantial body of literature on the creation and evaluation of judgmental forecasts, but few studies have been conducted in a retailing context (Lawrence, Goodwin, O'Connor, & Önkal, 2006), though one of the four companies studied by Fildes et al. (2009) was a non-food retail chain which adjusted some 20% of their forecasts. Overall, these adjusted retail forecasts were biased and inefficient, with no value being added by the adjustment process. One possible explanation for this may be that, despite accuracy being the stated objective, many of the adjustments were motivated by stocking/service-level considerations. Some studies have shown that judgmental adjustments can enhance baseline forecasts during promotions, but not systematically; instead, more advanced statistical models that include promotional indicators have proved better than the expert adjustments (Lim & O'Connor, 1996; Trapero et al., 2014; Trapero, Pedregal, Fildes, & Kourentzes, 2013), though the evidence is from manufacturing. Judgmental adjustments are often applied in retailing, as we discuss in Section 7; nevertheless, the scale of retail forecasting necessitates a more selective approach than when considering adjustments in manufacturing (for the vignettes described in Appendix B, the figures varied from 0% online to 40%, compared to around 70% for manufacturers). This naturally motivates the search for modeling methods that include promotional and other variables. No evidence has been collected from the field as to how the inclusion of such explanatory variables affects the adjustment process, but case vignette 1 in Appendix B demonstrates that the use of an econometric modeling approach does not preclude subsequent judgmental adjustment. The experimental evidence suggests that adjustments can partially take into account causal information that is already embedded in the model, though adjustments continue to be made – which, as expected, are smaller than optimal (Lim & O'Connor, 1996; Sroginis, Fildes, & Kourentzes, 2018). The research question here is whether software can be designed to ensure that expert information is incorporated into the forecast (through demand planning meetings, for example), so as to avoid double counting and exclude the irrelevant cues that commonly form part of the forecasting support system (FSS) and the associated organizational process.

5.4. Linear causal methods

Another stream of studies uses a model-based system to forecast product sales by taking into account promotional (and other) information directly. Usually, these methods are based on multiple linear regression models or more complex causal models, the exogenous inputs of which correspond to the factors discussed in Section 4.3, especially seasonality, calendar events, weather conditions, price, and promotion features.

The merit of linear regression is that it is simple, easy and fast to fit, meaning that it is feasible for large-scale product-level forecasting problems. A variety of forecasting solutions have been based on regression models with different specifications; multiplicative (log–log), exponential (semi-log) and log-reciprocal functional forms are the parametric specifications that are used most widely for representing nonlinearities in the sales response to promotional instruments. A well-known example is the SCAN*PRO model and its extensions, which decomposes the sales for a brand into own- and cross-brand effects of prices, feature advertising, aisle displays, week effects, and store effects (Andrews et al., 2008; Foekens et al., 1994; Van Heerde, Leeflang, & Wittink, 2000, 2001). PromoCast (Cooper et al., 1999) is another well-known promotion-event forecasting model; it uses a static cross-sectional regression analysis of SKU $\times$ store sales under a variety of promotion conditions, with store- and chain-specific historical performance information. Divakar et al. (2005) also employ a dynamic regression model that captures the effects of variables such as past sales, trend, own and competitor prices and promotional variables, as well as seasonality, in a regional analysis of the cola market.

In addition to linear regressions, more sophisticated models that include complex error correlation structures have also been proposed. Curry, Divakar, Mathur, and Whiteman (1995) propose a Bayesian VAR model for forecasting canned soup product sales at the brand level. The model includes the sales, price, and advertisement of four competing brands as endogenous variables. Baltas (2005) proposes a panel regression model that admits store heterogeneity, periodic sales variation, chain-wide sales shocks, and sales dynamics. Recently, Huang et al. (2014) and Ma et al. (2016) used the IRI data set (Bronnenberg et al., 2008) to develop autoregressive distributed lag (ADL) models and evaluated them on SKU data for many categories and a number of stores, with the latter study showing that a Lasso procedure could take into account successfully both intra-category promotional variables (12% improvement in MAE) and inter-category promotion effects (worth a further 0.6%). Arunraj and Ahrens (2015) developed a seasonal autoregressive integrated moving average with external variables (SARIMAX) model for forecasting the daily sales of bananas in a German retail store. Michis (2015) proposed a wavelet smoothing method for improving the conditional forecasts generated from linear regression sales response models.

With such a variety of models, the key question is: what is known about their relative performance? We consider this in Section 5.6 and Table A.1.

5.5. Nonlinear and machine learning methods

Nonlinear methods include traditional nonlinear regressions, non- or semi-parametric regressions, and fuzzy and machine learning algorithms. Unlike linear regression models, nonlinear methods allow the use of arbitrary nonlinear approximation functions that are derived (learned) from the data directly, and this increased generality improves the potential to provide more accurate forecasts (though with an increased danger of overfitting).

For grocery products, most of the published research has found improvements in forecasting accuracy from using nonlinear models rather than linear regressions. The models used include back-propagation neural networks (Aburto & Weber, 2007; Ainscough & Aronson, 1999), fuzzy neural networks (Kuo, 2001), regression trees (Gür Ali et al., 2009), gray relation analysis and multilayer functional link networks (Chen & Ou, 2009, 2011), two-level switching models that select between a simple moving average and a nonlinear predictor (e.g., *k*-nearest neighbor, decision trees) based on the characteristics of the time series (Žliobaitė, Bakker, & Pechenizkiy, 2012), support vector machines (Pillo, Latorre, Lucidi, & Procacci, 2016), wavelets (Michis, 2015; da Veiga, da Veiga, Puchalski, dos Santos Coelho, & Tortato, 2016), a fuzzy set approach (da Veiga et al., 2016), and Bayesian P-splines (Lang et al., 2015). One exception where nonlinearities led to a poor performance is the study by van Donselaar et al. (2016), who analyzed the impacts of relative price discounts on product sales during a promotion but did not find conclusive evidence for the presence of threshold and/or saturation levels for price discounts for perishable products. Despite the enthusiastic examination of this wide range of ML approaches, if nonlinearities were commonplace and easy to identify, we would expect to see such models being used more in practice. However, we do see machine learning methods becoming more deeply embedded in commercial software (see for example JDA's purchase of BlueYonder in 2018).

For apparel or fashion products, studies that use nonlinear models have been found consistently to provide better forecasts. Choi, Hui, Liu, Ng, and Yu (2014) argued that this is due to the demand for apparel and fashion products being notoriously highly volatile, thus making it difficult to identify the underlying pattern. Hence, the well-established and traditional statistical methods fail to make sound predictions. Of these techniques, neural networks (NN) are probably the most studied. For example, when searching for the ideal neural network structure for an apparel sales forecasting system, Au, Choi, and Yu (2008) proposed an evolutionary computational approach which they found to be more accurate than a traditional SARIMA model. However, standard gradient learning algorithm estimation such as back-propagation (BP) NN is relatively more time-consuming. Extreme learning machines (ELM), which provide much faster learning speeds, have been adopted in a number of fashion forecasting studies (Wong & Guo, 2010; Xia, Zhang, Weng, & Ye, 2012; Yu, Choi, & Hui, 2011). The experimental results have

shown that the performance of the ELM is more effective than those of traditional BPNN models for fashion sales forecasting, but in practice, their accuracy compared to BPNN is moot at best. When the historical data are limited, it is claimed that grey model based methods perform better (Choi et al., 2014; Xia & Wong, 2014); however, the results need to be validated further based on a larger sample of series. Du, Leung, and Kwong (2015) proposed a multi-objective optimization-based neural network (MOONN) model which they claimed was superior to several of the above-mentioned methods for the short-term replenishment forecasting problem.

The scalability of nonlinear models is usually poor, so the amount of training time required to build and maintain nonlinear forecasting models for tens of thousands of SKUs in hundreds of stores becomes a serious concern (though parallel computing with modern large computer clusters could help, see Seeger et al., 2016, as an example). Note, though, that using a trained model to produce forecasts is much faster typically than training the model in the first place. The size of the pooled dataset and memory limitations also raise estimation problems. For these reasons, researchers who have tried to test the superiority of nonlinear models at the product level usually work at a very small scale (i.e., tens of items).

An exception to the small scale studies is that of Mukherjee, Shankar, Ghosh, Tathawadekar, and Chaudhury (2018), who developed a demand density forecasting model with a multi-layered hybrid deep neural network for inventory management in Flipkart.com. They modeled the demand density as a learned mixture of Gaussian distributions, which combined both time series and associative factors. They experimented on a dataset of over ten thousand products from Flipkart and showed significant improvements in forecasting accuracy over the Cubist (a hybrid of decision trees and linear regression models).

A further limitation of ML methods is that they cannot automatically produce prediction intervals: a necessary requirement for inventory decisions, as we argued in Section 5.1. However, Flunkert, Salinas, and Gasthaus (2017) proposed an autoregressive recurrent network model, named DeepAR, which could produce accurate probabilistic forecasts directly by outputting the parameters of a negative binomial. Using an alternative approach, Wen, Torkkola, Narayanaswamy, and Madeka (2017) proposed a sequence-to-sequence neural network that generates multi-horizon quantile forecasts. Both studies applied their models to predicting the future demand for items sold on Amazon.com, and showed promising results compared with a number of variants; however, their models were not compared with state-of-the-art benchmarks.

We conclude that the evidence for nonlinearity generally leading to better forecasting accuracy is weak, with the positive evidence probably arising from a publication bias, which may be amplified by lack of knowledge about good forecast evaluation practice in the machine learning community, along with a certain amount of hype. The studies cited have many limitations, which are summarized in the next section.

5.6. Comparative evaluation

Table A.1 in Appendix A summarizes the studies described in the previous sections, concentrating on how thoroughly they have been evaluated. We begin by describing the focus of each study and the range of products that it analyzed (in columns 2 and 3). There are established guidelines for the rigorous evaluation of forecasting methods (Tashman, 2000), and these are considered in column 4; in particular, the comparison of the proposed model with simple benchmarks (column 5). The authors' conclusions are summarized in the final column. The table is necessarily selective, and includes only studies that use retail product-level data in their empirical tests and are novel enough or involve sufficiently extensive testing to justify their discussion in the previous subsections. A few papers which did not report their experimental settings (e.g., horizon, lead time, or validation sets) clearly have been omitted.

As Table A.1 makes clear, we face limited evidence. The studies summarized therein have little in common methodologically, and most suffer from a lack of generalizability. Many are applied to too few SKUs and do not include standard forecasting methods using a well-defined out-of-sample testing procedure over a specified forecast horizon. However, there are some studies that consider a wide range of SKUs, sometimes at the store level, and we focus primarily on the studies which transcend these concerns.

5.6.1. Univariate benchmarks

There are few comparative examinations of univariate methods applied to product-level retail data where the datasets (as we have described them) display high volatility and intra-week seasonality, with few series showing any strong signs of a trend over the relatively short forecast horizons. Taylor (2007) examined the daily SKU demand for 256 products at a single supermarket, excluding all slow (intermittent) items. His proposed novel method of exponentially-weighted quantile regression outperformed both the exponential smoothing benchmarks and the optimized company procedure. Ramos et al. (2015) compared state space and ARIMA models when forecasting the monthly demand for women's footwear for lead times of 1 to 12, with little difference in accuracy. However, such aggregate forecasts are of little value to the retailer. Ramos and Fildes (2017) extended the analysis to 988 SKUs in 203 categories, considering a wide range of univariate methods, again excluding intermittent data, with TBATS (trigonometric, Box–Cox transform, ARMA errors, trend, and seasonal components) being the best performer. Here, the characteristics of the data were similar to those of Taylor's (2007) data: highly volatile at the store level. Standard univariate methods such as exponential smoothing (ETS, i.e., the state space formulation of exponential smoothing methods; Hyndman et al., 2008) do not perform well (e.g. Ma et al., 2016, who compared ETS, the error-trend-seasonality formulation, with causal methods).

Retailers' vast numbers of intermittent series (535k from Amazon were used by Seeger et al., 2016) require new methods to be tailored to these data characteristics and computational constraints (Flunkert et al., 2017; Seeger et al., 2016). While the Bayesian methods developed by Seeger et al. were shown to outperform a state space method designed for intermittent data (Snyder, Ord, & Beaumont, 2012), they have not been compared to standard intermittent benchmarks, with Flunkert et al.'s (2017) study of autoregressive recurrent neural networks eschewing promotional data. Nevertheless, they have shown considerable promise.

Research into comparative univariate methods now needs to concentrate on the structural characteristics of retail data: general studies such as the M4 competition (Makridakis, Spiliotis, & Assimakopoulos, 2018) tell us little that is of direct relevance to the retailer. Instead, what such competitions and the various commentaries have done is to develop a methodology for carrying out such comparisons (Fildes & Ord, 2002).

5.6.2. Multivariate methods

There are many studies that inform the development of multivariate forecasting methods. Key issues are the variables to be considered for inclusion (e.g. intra- and inter-category promotional variables, Ma et al., 2016; or online consumer reviews, Schneider & Gupta, 2016), and the level of pooling to be used to estimate the elasticities, with a trade-off between sample size and heterogeneity in the model. As we have just discussed, researchers have proposed various novel machine learning methods such regression trees and neural networks and their variants, which they claim are successful; however, the evaluations conducted are too limited. In contrast, Gür Ali et al. (2009) compared a number of machine learning methods to the base-lift benchmark and evaluated them over many store $\times$ SKU combinations. Huang et al. (2014) and Ma et al. (2016) compared the ADL model forecasts for various lead times and loss functions with simple smoothing models updated for promotional periods using the last promotional uplift, and found that the causal models performed best for a range of categories. Often, detailed promotional data are unavailable and simpler models must be used, so Ramos and Fildes (2017) conducted an extensive evaluation (988 SKUs in 203 categories) and showed that various multivariate methods, extended by including the exogenous price information, outperform the univariate methods, with typical gains of above 10%. For promotional periods, the gains are typically higher: in fact, the Lasso-based ADL models of Ma et al. (2016) continued to outperform ETS substantially even for non-promotional periods (for the focal product).

Overall, the multivariate studies show substantial accuracy improvements for SKU-level forecasts over univariate benchmarks. However, while recent studies have included a wide range of categories, the studies have focused mostly on groceries, neglecting other product groups. Intermittency and the effects of market instability are yet to be explored fully. Wide-ranging evidence of the benefits of machine learning algorithms is needed if we are to believe the hype that both researchers and software companies have generated.

6. New product demand forecasting in retail

Forecasting the demand for new products is a more difficult task than forecasting for existing products, because of the lack of direct historical product data. However, such an effort is essential despite the complexity of the task and its relatively low accuracy (Brown & Hunter, 1967; Kahn, 2002), as it drives a variety of multifunctional decisions. These include purchasing, inventory levels, decisions related to logistics, the effect on the overall assortment's profitability, and financial expectations for the new product. The general literature on this topic is vast and even by the mid-1980s, Assmus (1984) found the number of methods too numerous to include in his review paper. Since then, many new product forecasting methods (and models) have been developed, and a number of recent reviews on this topic have been made from different perspectives though none focus any attention on the retail new product decision (Chandrasekaran & Tellis, 2007; Goodwin, Meeran, & Dyussekeneva, 2014; Machuca, Sainz, & Costa, 2014; Meade & Islam, 2006). In general, new product sales forecasting methods can be grouped into three broad categories: (i) the judgmental approach, which involves management judgment based in part on past experience; (ii) the market research approach, where survey data are used to forecast customers' purchasing potential; and (iii) the analogical approach, where the forecaster assumes that the product will behave as "comparable products" have behaved, a comparison which entails the identification of such comparators and itself is often heavily judgmental. These various techniques for new product forecasting are described by Ord, Fildes, and Kourentzes (2017), for example. Past survey evidence shows that customer/market research, followed by management judgment, were the methods most commonly used (Gartner & Thomas, 1993; Kahn, 2002; Kahn & Chase, 2018). Judgment was particularly important where sales of hi-tech products were being forecast and the novelty of the product was associated with a greater uncertainty in relation to sales (Lynn, Schnaars, & Skov, 1999). However, note that all of these surveys apparently addressed only manufacturers, and not necessarily consumer product manufacturers: there do not seem to be any survey data on new product forecasting practices in retail or consumer products specifically, though practices here are likely to be similar to those in other industries.

The forecasting of new products by retailers has some of the same general characteristics as the generic problem, but also poses some specific issues. When the retailer also acts as designer and underwriter, the forecasting problem faced is the same as at any manufacturer, and we do not discuss this further. However, when there is a decision to be made about adding a product to an existing category, there is a requirement to forecast the overall demand across stores (for a decision-relevant time horizon), the cumulative purchase-repeat purchase path,

and the cannibalization effects on other products in the category. In all likelihood, the addition of a new product would be expected to grow the category as well, and in fact to increase the likelihood of a product being accepted into the retailer's portfolio (van Everdingen, Sloot, van Nierop, & Verhoef, 2011). The decision to take on the product then depends on the retailer's demand forecasts (of SKU, category), the profit forecasts derived from the demand forecasts, and the support offered to the retailer by the manufacturer (which itself influences demand, e.g. Ze & Bell, 2003). Fisher and Vaidyanathan (2014) proposed a method for estimating intra-assortment substitution effects and used it to forecast an entire new assortment.

With no direct historical data, the formal approach that is adopted most often (we speculate) is the use of analogy. This approach involves forecasting new products by making use of the past histories of similar products (Gilliland & Guseman, 2009). There are two variations:

(1) Models can be fitted to the historical data of the analogs and used to produce forecasts for the new product on the assumption that its adoption will follow a similar time series pattern (Bayus, 1993; Hu, Acimovic, Erize, Thomas, & Mieghem, 2019). In a Bayesian framework, this can be formalized so that the predecessor's history is used to provide initial priors for the new product and then these priors are updated as the new product's sales are observed (cf., Lenk & Rao, 1990).

(2) Alternatively, especially if the new product is replacing another similar product, one can use the history of the previous item, which is typically carried over in the retailer's database.

If there is no direct substitute, then one approach which recognizes the distinction is to identify a set of features and attributes of the product that are similar to those of existing products, such as brand, flavor, color, pattern, price, and its target customer segments, and then to group the demand characteristics from those items so as to forecast the demand for the new product. Ferreira et al. (2015) applied this approach to an online fashion retailer; Schneider and Gupta (2016) applied it to online Amazon sales of tablet computers; Tanaka (2010) applied it to data on early sales of books and consumer electronics for forecasting longer-term 6-month sales; and Wright and Stern (2015) used analogs combined with trial data for various consumer products. Unfortunately, the evaluation of these models typically has not focused on the retailer's decision requirements – whether or not to stock, and how much to order. The identification of appropriate analogies is also difficult, as Goodwin, Dyussekeneva, and Meeran (2013) showed (though not in a retail context); however, it may be that the plethora of examples in retail and fashion has made the methods more successful, as Thomassey (2014) claimed and as Ferreira et al. (2015) illustrated in a study of an online fashion retailer.

In recent years, researchers have argued that web-based analytics are useful for predicting the potential performance of a product launch. This opens a new direction for market-research-based new product forecasting. For example, Xiong and Bharadwaj (2014) found that the shape of the curve of the prerelease buzz evolution trajectory generated from online blog and forum postings adds power for predicting the new product performance of video game sales. Schneider and Gupta (2016) used text mining of consumer reviews of new Amazon products, and their incorporation into the model improved the accuracy one week ahead. A considerable amount of research has been published on this topic recently, but most of it neglects a retail perspective, and, as Schaer et al. (2019) show in their literature review and empirical work (on computer games), the predictive value, if any, is only short-term (e.g. up to a month).

Retailers sometimes use test markets before deciding on a national launch. Past research has been concerned mainly with the optimal selection of the sample markets (Mostard, Teunter, & de Koster, 2011) and test-market-based forecasting models. For instance, Fisher and Rajaram (2000) presented a clustering and linear programming based methodology for both selecting stores at which to conduct the test and creating a forecast for the forthcoming season for the whole chain based on test results. In a contrasting approach, Wright and Stern (2015) showed how a simple model of trial results based on analagous products from the first 13 weeks could be used for other new products and considered in the national launch decision.

Fashion products provide a particular new product forecasting problem (Gallien, Mersereau, Garro, Mora, & Vidal, 2015). Before each season, there is a need for a forecast of total product sales, whether distributed via the high-street or online, and lead times may be lengthy. For many years, a variant of the problem arose through the use of mail-order catalogs (see e.g. Mostard et al., 2011), though the increased prevalence of online has left the catalog as merely an adjunct to online marketing.

Typically, analogous products are used, though more advanced methods can incorporate product attributes as well as their sales. These attributes are used to cluster past products in order to develop sales profiles for new products (Thomassey & Fiordaliso, 2006), with Baardman, Levin, Perakis, and Singhvi (2018) developing a joint approach of clustering and forecasting which they show to provide improvements over the sequential approach. A Bayesian framework can incorporate various data sources prior to the product launch, and those predictions are updated later as new data become available. For example, Yelland and Dong (2014) derived priors from past products, and used the early sales data on the new product to update the model parameters once they were available. However, early sales are only potentially helpful for the late season. Given that judgment is the predominant approach, the question arises as to what sort of FSS might improve on unaided judgment. In a study of pop music charts, Seifert, Siemsen, Hadida, and Eisingerich (2015) concluded that the provision of contextual information (such as the promotional campaign and critical reviews) and the provision of historical information on previous success both improved the accuracy, though the value of each depended on what model-based information was provided through the FSS.

In summary, new product demand forecasts are an important aspect of retail forecasting. Apart from the fashion

industry and, more recently, online businesses, research has been limited. Standard approaches are claimed to apply, and analogies are at the heart of what formal methods are used. Some limited survey evidence has been collected for examining the methods that are in use. Judgment influenced by analogies was found to be the favored approach. The effect of introducing a new product into a category is unlikely to be modeled formally, whether or not it is recognized, although software has been developed that attempts this (e.g. SAS, which includes text mining and sentiment analysis). Finally, the selection bias noted in Section 3.3 should be kept in mind when using data from historical new product introductions.

7. Retail forecasting practice

No recent surveys of forecasting practice in retail have been carried out since Peterson (1993), who found a limited use of econometric (causal) methods: managerial judgment was used the most, followed by simple univariate methods. Apparently, test markets and surveys were also popular. McCarthy, Davis, Golicic, and Mentzer (2006) reached much the same conclusion in a general survey that included retail. A more tangential approach is to examine what software suppliers offer. The largest software providers, such as SAP and SAS, offer a full range of forecasting methods in their demand planning suites, starting with simple univariate methods, but more advanced multivariate models are also available in modules that either are additional to the popular base suite (at an extra cost) or require tailoring to the company's database and cannot be used automatically. For example, SAP now offer the retail-specific customer activity repository (CAR), though APO or other software providing only univariate procedures remained common as of 2019. SAS Forecast Server and Visual Forecasting (open source) automatically uses all categories of modeling, including multivariate methods (e.g. ARIMAX, dynamic regression, unobserved component models) across a company's entire product portfolio. Specialist providers such as Relex may perhaps provide a pointer as to how retail practice is expected to change: they add machine learning into the mix of regression-based methods, and its inclusion is becoming an industry standard (though, as we have noted from the empirical comparisons, it is far from clear quite what it adds to the suite of available methods). However, typically changes in practice are slow. For instance, collaboration between retailers and suppliers has long been a hot topic for both academics and software providers, but Weller and Crone (2012) showed, through a rigorous survey, how little this innovative sharing of retailers' EPOS data and forecasts has impacted on manufacturing practice, and such collaborations can be fraught with difficulty. Nor was it seen as a major opportunity in the vignettes of practice described below.

A second aspect of the diffusion of new modeling practices into industry is the need for trained staff. While packages such as SAP's CAR and methods such as those proposed by Ma et al. (2016) attempt to make modeling automatic, there remains substantial doubt as to how acceptable such forecasts are in organizational practice. Without staff who are trained in regression-based methods, the likelihood is that such methods will not be accepted fully as a basis for the decisions that need to be made: judgmental overrides remain common from operational demand forecasts to the tactical and strategic (such as site location), where the evidence is that, while models are used, expert judgment remains a major component (Wood & Reynolds, 2013). The stylized cases that we present in Appendix B covering demand planning activities show that judgment remains a key feature, even when the model base in the FSS includes many causal drivers (e.g. promotional types, holidays). Based on the experimental evidence, Sroginis et al. (2018) showed that model- based causal forecasts were not fully responded to and judgment still remained an important feature and this finding was reflected in retailer practice.

In order to understand more as to how the alternative approaches have been implemented and the challenges faced we describe briefly five archetypal chain-store forecasting organizations, based on detailed discussions. We also use the format to discuss additional issues we have identified in interviews with a further three companies. A summary is presented in Appendix B. Additional evidence can be found in presentations by Seaman (2018) relating to Walmart and by Januschowski (2017) for Amazon. Both presentations represent data science specialist support and therefore focus more on algorithms. However, the scale of both operations is a constraining factor on the algorithms that can be adopted (in Walmart, a dynamic local linear seasonal model, in Amazon, this includes an autoregressive neural net).

These vignettes capturing practice demonstrate a number of key points:

(i) Despite the availability of commercial software including sophisticated causal modeling and nonlinear methods (sometimes included with the ill-defined term, 'demand sensing'), uptake of such advanced tools has been uneven. It is not clear whether these tools offer advantages commensurate with the higher total cost of ownership in a real retail forecasting situation and few companies have routinized the use of these more advanced procedures; promotional modelling at SKU or SKU $\times$ store level remains simplistic with limited use of lags, promotional definitions and intracategory data.
(ii) New product forecasting remains heavily judgmental and informal.
(iii) Intermittent demand is a key problem where established research solutions have not been adopted.
(iv) KPIs and accuracy measurement are typically not given sufficient attention with data that may well be heavily influenced by intermittency and extremes. In most of our cases, MAPE is used which is particularly prone to distortion (as we discussed in Section 5.1). Volatility in updated forecasts and consequential orders are potentially important for supply chain planning (Seaman, 2018) but this is seldom part of the appraisal. Lead time issues linked

to the supply chain are rarely considered. Predictive densities and quantiles are rarely (never?) considered, despite their importance in ordering and inventory decisions.

(v) Demand planning in retailing is manpower intensive, and staff may have overly limited statistical knowledge. In a recent development some of the companies have introduced a separate data science group. There is some variability in the role, where in some examples top down processes have been established whilst others take more of a consulting role. In most retailers, demand forecasting is still overly influenced by IT constraints, with too little emphasis on forecasting knowledge.

(vi) Judgmental intervention superimposed on model based forecasts remains a significant element in retail forecasting beyond the requirements of new product forecasting. Adjustments are typically top-down from the demand planning team but, nevertheless, ordering decisions at the store may implicitly override the SKU × store forecasts with sub-optimal results (van Donselaar, Gaur, Van Woensel, Broekmeulen, & Fransoo, 2010). No examples of brick-and-mortar retailers were found that relied on the fully automatic use of software.

(vii) More tentatively (based on our convenience sample of retailers and software suppliers), the diffusion of best practice modelling remains slow, perhaps due to the installed base of legacy software and the rarity of formally trained forecasters, statisticians or data scientists among retailers.

8. Conclusions and future research

The retail sector is experiencing seismic change. At the strategic level, existing brick-and-mortar retailers face hard choices with regard to their stores and their embrace of online activities. The forecasts, naïve though they are, shown in Figs. 1 and 2 underline the continuing potential for disruption. The research literature has not been engaged with the questions posed by such rapid structural change in the development of online competition, with most though not all companies moving from presenting a chiefly a brick-and-mortar offering, possibly supplemented with catalog sales (primarily with a focus on fashion goods), to one where online has both superseded and expanded on the previous range of catalog offerings. Amazon looms over the whole sector, particularly in the US, where its purchase of Whole Foods and its move into physical stores has generated shockwaves. This has presented a variety of forecasting challenges. At one level, each online retailer has a developing time series history of products that can be used with standard forecasting methods. However, the addition of an online alternative channel by an established brick-and-mortar retailer raises fresh problems, and the attempt to make the customer view the alternatives as seamless ("Omnichannel" shopping) raises the question of complementarity versus substitution. The decision to launch an online service should be based on forecasts of total sales and profits for the two alternatives over the planning horizon. However, no studies examining this problem have been found.

A further question that is raised by the dramatic changes in retail shopping habits in some countries is the mix of store locations (mega-store out-of-town, supermarket and local convenience store). The established location model methods which previously had high credibility have been undermined by the change in consumer behavior, leaving a database that is devoid of any directly relevant data on the basis of which new models can be constructed.

While the strategic issues that face retailers have been largely neglected by researchers, the operational questions that relate to the identification of more effective forecasting methods have been resolved in part. First, and most importantly, the choice of benchmark is important even in non-promoted periods, partly of course because small improvements over many thousands of SKUs lead to major financial and service benefits. Whilst much of the earlier research was limited by its breadth, more recent work has considered a range of stores and many SKUs in different categories, thus helping with generalizations: conclusions rarely apply to all categories, whereas we speculate that there may be greater robustness across stores and SKUs within categories. Further research seeking explanations for differing relative performances would be interesting, linking technical forecasting issues to the characteristics of the market (see for example Huang et al., 2019). When promotional events are considered, again research has demonstrated the benefits of more complex modeling approaches, as was discussed in Section 5.6. The value of these more advanced models can be demonstrated by the added value that comes from the inclusion of additional classes of variable (e.g. model dynamics were worth an additional 3.42% in profit, see Ma & Fildes, 2017). They also can be used to help in understanding the importance of business pricing rules, whereas companies seem to still rely on ad-hoc procedures (Watson, Wood, & Fernie, 2015).

However, one field in which more research is needed urgently is the proliferating variety of promotions (see Fig. 9). It is no longer enough to consider "Feature type 1" and "Feature type 2" as in the IRI dataset; instead, promotions are characterized by a multitude of different (interacting) tactics. Also, promotions do not necessarily run for exactly one week that coincides with the calendar week. Rather, retailers start promotions on days with lower traffic – typically in the middle of the week – in order to lure customers to their stores. Thus, older datasets of a weekly granularity are of dubious usefulness. Researchers should aim to obtain more realistic datasets, and retailers should be open about providing them.

Quantile, density or volatility forecasting is very much an under-investigated aspect of retail forecasting, even among studies of SKU-level forecasting. More research in this direction, as well as on the connection to inventory control, would be desirable. It may be possible to apply models for safety stock determination from other industries, or to make use of volatility forecasting models such as (G)ARCH.

There remain operational forecasting challenges for omni-retailers, such as how the geographical mix of online sales affects the optimal warehousing and distribution network.

A developing opportunity that in fact researchers have embraced in part already, is the ready availability of 'big data' (Bradlow, Gangwar, Kopalle, & Voleti, 2017; Fisher & Raman, 2018) and its potential for improving demand forecasting: every kind of customer behavior, whether in-store or through online activity, can be integrated into SKU demand forecasts (Boone, Ganeshan, Jain, & Sanders, 2019). Some have claimed that this route is set to undermine more conventional demand forecasting by shifting the attention to the individual customer. However, as Kolassa (2017) pointed out, the technical hurdles of translating individual (as opposed to segmented) demand histories into aggregate SKU-level forecasts are yet to be overcome successfully.

A second marketing problem arises with online purchasing, where the web habits of customers, both actual and potential, can be analyzed and linked to purchasing through different search patterns (Jayasankaraprasad & Kathyayani, 2014). This leads to the question of how websites should be designed in order to maximize their effectiveness.

The question of whether the value of this behavioral data is a chimera or a research gap soon to be filled convincingly remains to be answered. The related issues of customer reviews, Google searches and other online forums and their impacts on sales are reviewed by Schaer et al. (2019), who demonstrated that much of the existing research has transgressed the principles of good forecasting research by failing to focus on out-of-sample comparative evaluations at decision-relevant lead times.

Online purchasing behaviors raise specific operational problems for the forecaster that are not relevant in brick-and-mortar alternatives, in particular returns and fraudulent purchasing behaviors, both of which show a high prevalence and are category-specific. The problems can be analyzed using data mining techniques where time series behavioral data are included.

As part of the research underpinning this article, we consulted with various retailers as to what they see as their major forecasting challenges. Despite the contrast between the methods that are used most commonly (exponential smoothing with promotional profiles and event dummies) and those that are likely to be most accurate, there was little dissatisfaction with the methods in use. Various outstanding methodological issues were noted: (i) the need for methods that are suited to intermittent demand, and (ii) scalability (see e.g. Januschowski, 2017; Seaman, 2018), where the algorithms often need to be run overnight for upward of 40K SKUs for Walmart, with many hundreds of stores, and upward of 500 K SKUs and multiple DCs for Amazon. As a consequence, computational resources are a limitation. To these comments we add: (iii) the increased use of AI/machine learning methods and the lack of validation in the research literature, and (iv) the use of judgment in many applications in which a retail sales and operations planning (S&OP: see Harwell, 2015) process aims to combine different sources of information, leading to judgmental interventions. The move towards machine learning methods raises two other significant issues: the users' need to understand the impacts of the different drivers that are included in the ML implementation and their ability to add value when making judgmental adjustments (see Section 5.3). Finally, a further issue that was highlighted by the retail supply chain experts we interviewed was the increasing importance of new products with short lifetime histories.

For researchers, a number of important problems remain under-researched in addition to the strategic, discussed earlier. The relationship between the SKU-level demand across channels for different categories is important as a practical problem, while the search for market-based explanations makes it an interesting theoretical research area too. The second operational problem relates to the development of methods for supporting the introduction of new products and the implications for the supply chain. As yet, there has been no work evaluating the comparative accuracies of the different approaches.

For software developers, the main problem raised by this review relates to the development of automatic scalable models that are robust to the data limitations that are common in retail operations. Issues of data hierarchies where research solutions exist have seen limited implementation. A second underdeveloped area is the provision of error statistics and the quantiles that are needed for replenishment: forecast error data over lead times are rarely accessible.

Finally, with increasingly vast quantities of data linked to the 'big data' generated by observed consumer behavior, we expect retail forecasting to provide a test-bed for the integration of micro-data into more aggregate demand forecasts.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

A.1. Product sales forecasting papers

See Table A.1.

Table A.1
A brief summary of representative research papers on product-level sales forecasting. The papers, discussed in the text, have been selected as those that contain a wider range of comparisons and more complete evaluations. (Blattberg & George, 1991; Kumar & Patel, 2010; Lu, 2014; Ren, Choi, & Liu, 2015).

(1) Reference	(2) Focus	(3) Data: range and granularity in the product, location and time dimensions	(4) Variables	(5) Forecast horizon and evaluation	(6) Baseline methods	(7) Conclusion and notes
Blattberg and George (1991)	Shrinkage estimation of price and promotional elasticities in pooled data; focus is on "more reasonable coefficient estimates" rather than on better forecasts	3 grocery chains, 4 brands of bathroom tissue; 110, 112 & 126 weeks of data for the 3 chains	Brand volumes, relative prices to average competitive prices, discount, display, feature, max competitive discount	Final 10%, middle 20% and random 20% of the samples, single origin; relative MSE	Semi-log regression	Pooling brands that have different pricing philosophies yielded more reasonable parameter estimates and improved prediction. Predictive improvement slight for only one pooling alternative. Poor design.
Foekens et al. (1994)	Compare pooling across chains and stores, including the homogeneous SCAN*PRO model of retail promotion effects at different levels of aggregation and pooling	3 brands of food products sold in 40 stores from 3 chains, 104 weeks	Sales volume; own and cross prices, own and cross displays and features	52 weeks for estimation, single origin up-to-4-week, up-to-12-week and up-to-52-week forecasts; MAPE, MdRAE	Models with heterogeneous response parameters across retail chains	A heterogeneous store model provided a superior accuracy at the chain and market levels.
Curry et al. (1995)	Forecasting Bayesian vector autoregression modeling promotional activity	4 brands in one category (canned soup) at the market level (aggregated over 40 stores), 124 weeks	SKU aggregated sales volume: prices, display, feature, advertising exogenous; seasonal indices	Rolling-origin 1–16-week-ahead forecasts aggregated in Theil's U	Univariate and multivariate time series models: BVAR, MARMA, ARIMA, exponential smoothing	BVAR provided more accurate forecasts; MARMA poor, ARIMA competitive.
Cooper et al. (1999)	The implementation of a promotion event forecasting system, PromoCast, and its performances in several pilot applications and validity studies	Two analyses: (1) unspecified number of SKUs in 311 stores, (2) 212 fast-moving SKUs in 178 stores; time granularity: promotional events spanning one or multiple weeks	70 variables capture the effects of promotion style, SKU and store promotion history, and seasonality	Two longitudinal cross-validations with 20,710 and 30,569 promotion events respectively, horizon not reported; evaluation on errors in replenishment cases of typically 12 units	Historical averages; base lift	A regression-style promotion model helped to improve forecasts in large-scale applications; store-by-store forecasts can help retailers with problems of running out of stock or overstocking. Much detail on promotional events.
Ainscough and Aronson (1999)	Examines neural networks as an alternative to traditional statistical methods for the analysis of scanner data	A national brand of strawberry yogurt, 575 weeks of sales data are aggregated from 6 stores	Lagged sales; price, display, feature, and their two-way interactions	20% of the sample as the validation set (approx. 20 weeks); MSE	OLS regression	It was found that three- and four-layer neural networks yielded significantly better predictions than OLS regression.
Cooper and Giuffrida (2000)	Examines the value of residual mining on SKU-level promotional forecasting	1.6 million grocery SKU records from 96 outlets; focusing on sales forecasting during promotion events that may have various time lengths	Mining with nominal variables, such as manufactures, retailer stores or geographic areas	A holdout sample of 460,000 records; MAE	PromoCast forecast module	Mining residuals from a market response model reduced the promotional forecast errors.
Dekker et al. (2004)	Using aggregation and combined forecasting to improve the seasonal demand forecasts for groups of seasonal products	56 products from three product classes with 5 years of weekly sales data	None	Rolling holdout sample over 52 weeks, lead one week; sMAPE, MAD and MSE	SES; HW	Seasonal indexes that were estimated from the aggregation of similar seasonal time series proved helpful in improving the forecasting accuracy.
Trusov et al. (2006)	Evaluates the value of residual mining on SKU-level promotional forecasting with a focus on promotional adjustment; similar to Cooper & Giuffrida (2000)	8,195 records from the output of the PromoCast Forecast module	Categorical variables which describe product, store, or promotion conditions	Forecasting horizon is unclear; 4,000 records as the validation of correct promotional adjustment	PromoCast forecast module	Data mining the residuals from response modeling with product attributes improved the forecasts.
Thomassey and Fiordaliso (2006)	Short-term sales forecasting for numerous new items	482 textile items with 52 weeks of sales history	Sales histories are used to cluster items with similar patterns; product attributes are used to classify a new product to a cluster	285 of 482 used as test items; horizons of 1–50 weeks; RMSE, MAPE, MdAPE	Average sales of the items that belong to a given family	Clustering techniques and decision tree classifiers were useful for forecasting sales of new items with limited data.
Taylor (2007)	Forecasting daily supermarket sales using exponentially weighted quantile regression	256 time series of daily sales with median daily sales greater than or equal to 5, in length from 72 to 1436 observations	None	Horizon of 1 to 14 days; 42,633 post-sample sales observations; relative MAE, and coverage measures for prediction intervals	Univariate models (Naïve, Holt's, HW, etc.) and company procedure	Exponentially weighted quantile regression (EWQR) compared favorably with a variety of other univariate methods. Evaluates robust adaptations of EWQR.
Gür Ali et al. (2009)	SKU demand forecasting in the presence of promotions	168 store–SKU combinations from 4 categories of non-perishable items spanning a period of 76 weeks	100 features extracted from historical statistics of the past 4–12 weeks for capturing category, store, SKU, and promotion characteristics	Unclear forecasting horizon, 25 weeks of hold-out; only MAE used	Regression and base-lift model	Using more detailed input data was beneficial only if more advanced techniques were used; calls for incorporating SKU data from multiple stores and multiple subcategories into the same model. ES most accurate outside promotions.
Andrews et al. (2008)	Empirical comparisons of SCAN□PRO models with continuous and discrete representations of heterogeneity when forecasting store–brand sales	5 brands of shampoo from 28 stores, 109 weeks	Price, display, feature and their cross effects	Horizons of 1–28 weeks; RMSE	Homogeneous SCAN□PRO model	Accommodating store-level heterogeneity did not improve the accuracy of marketing mix elasticities relative to the homogeneous SCAN□PRO model.
Kumar and Patel (2010)	A new method of combining forecasts using clustering: clusters of items are identified based on the similarity of their sales forecasts, then a common forecast is computed for each cluster of items	411 items of women's summer clothing; 21 weeks of data for the selling season	None – price is constant	Rolling from week 5 to week 21, lead 1; average MSE	Individual forecasts based on ES, ARIMA, MA; results of individual methods not discussed	Combining forecasts from multiple items using clustering produced better sales forecasts than the individual forecasts.
Gür Ali (2013)	Driver moderator method in SKU–store-level sales prediction	Two datasets: (i) 451 SKU–store combinations of daily sales over two years; (ii) weekly data on 1020 SKUs from the IRI dataset over 4 years	Hundreds of features consisting of drivers, their interactions with modifiers, and dummies for months and days	No horizon is identified: one-fifth of the observations were chosen randomly as validation dataset; MAE and MASE	Neural network, OLS regression, and regression tree	Driver moderator model with pooled information across SKUs and stores provided more accurate forecasts.
Huang et al. (2014)	The value of competitive information in forecasting FMCG retail product sales	122 SKU aggregated across 83 FMCG stores in 200 weeks	Promotion on focal SKU, price diffusion index, promotional diffusion index, week dummies, calendar events, sales lags	Rolling out of sample over 70 forecast origins, leads 1,4 and 12; MAE, MASE, SMAPE, MAPE, AvgRelMAE	ADL with/without promotion information, univariate model, and base-lift model	Models that integrate competitive explanatory variables with the right variable selection process generated substantially more accurate forecasts.

(*continued on next page*)

Table A.1 (*continued*).

Lu (2014)	Selecting appropriate predictor variables and constructing effective forecasting models for computer products	5 computer products, 247 weeks, from a computer product retailer	A variety of variables constructed from the sales history, including lags, trends, growth ratios and volatilities	49 weeks of holdout; single origin; MAD, RMSE, MAPE, RMSPE	SVR, MARS, ARIMA	The hybrid model which uses MARS to select important forecasting variables and then inputs selected variables to SVR produced a better accuracy.
Voleti et al. (2015)	To assess the respective contributions of inter-product competition and brand–SKU hierarchy effects to explaining and predicting demand	96 SKUs under 15 brands, beer category data from 56 stores of a mid-sized grocery chain, 23 weeks	Inter-product competitive effects, marketing mix, month dummies	6 weeks of holdout, single origin, leads 1; RMSE	Standard log-log regression	SKU competition was more local than global, in that only subsets of products compete within groups of comparable products.
Lang et al. (2015)	Accounting simultaneously for heterogeneity and functional flexibility in store sales models	89 weeks of data on 8 brands in 81 stores from a major supermarket chain	Price, lowest price of a competing national brand, price of a private label brand, display, store-specific random slopes	9-fold cross-validation with 89 weeks data; ARMSE	Parametric/nonparametric regression models with/without considering store heterogeneity	Allowing for heterogeneity in addition to functional flexibility improved the predictive performance of a store sales model.
Arunraj and Ahrens (2015)	Interval prediction of perishable food daily sales	5 years of daily sales of bananas, measured in kilograms, from a typical food retail store	Promotion, discount, weather, weekday, month, holidays	Holdout 30% of all the observations; single origin, lead 1 horizon; MAPE, RMSE, percent of observations fall in 95% prediction intervals	SARIMA, SARIMAX, MLPNN, hybrid SARIMA and MLR	The proposed hybrid SARIMA and quantile regression provided better prediction intervals.
Ramos et al. (2015)	Performance comparison between state space and ARIMA models for consumer retail sales forecasting	Monthly sales of 5 categories of women's footwear, 64 observations	None	Rolling 1- to 12-month origins, leads 1–12; ME, MAE, RMSE, MPE, MAPE	ARIMA	State space and ARIMA produced similar forecasts.
Michis (2015)	Wavelet smoothing method of conditional sales forecasting	1 category, 1 brand, and 2 SKUs of detergents at the national level, 120 weekly observations for each	Weekly and monthly indicators, lagged sales, trend, holidays, average price per unit, total number of promotional packages	30 and 40 weeks as holdout; MSFE, FESD	Conditional regression without smoothing, with the moving-average or Baxter–King filters	Wavelet smoothing provided the best results when applied to highly volatile marketing time series.
Chern et al. (2015)	How electronic word-of-mouth affects product sales	52 best-selling cosmetic products, weekly , aggregated over 355 chain stores, 80 weeks	Strongly positive reviews, strongly negative reviews, neutral reviews, sales lag, time	Unclear how the authors set the validation period; MAPE	Moving average forecasts	Products with abundant online reviews obtained better forecasts using the proposed online word-of-mouth-based sales forecasting method.
Ren et al. (2015)	Fashion products replenishment forecasting	6 fashion items with 7 kinds of color, 36 weeks of data from a fashion boutique	Price, sales lags	12 weeks as holdout, leads 1; MSE, sMAPE	ARIMA, RVM, panel regression	Proposed panel data-based particle-filter (PDPF) model proved suitable for conducting fashion sales forecasting with limited data.
Du et al. (2015)	Fashion products replenishment forecasting	7 categories of fashion product sold in 3 cities, 17–48 samples for each product, the time dimension is unclear	3 early sales volumes, 1 weather index and 2 economic indices	4–12 samples as holdout for each product; RMSE, MAE, MAPE	Extreme learning machine and two variants	A non-dominated sorting adaptive differential evolution algorithm (NSJADE) to optimize the weights of neural networks (NNs) provided superior forecasting performance.
van Donselaar et al. (2016)	Forecasting promotional demand for perishable products	407 SKUs, 86 weeks of data from 4 perishable food categories, aggregated on a national level from a retail chain	Discount levels, weight of selling unit, shelf life, number of items in promotion, regular price, display, flyer, holiday, baseline sales	Validation on last 24 weeks; single origin; RMSE, MAPE, average bias	Regression models with 5 price discount dummies, linear discount, or quadratic discount; poor benchmark used	Modeling threshold and saturation effects led to worse forecasting performance than modeling price reductions linearly or quadratically for perishable products.
Gür Ali and Pinar (2016)	The value of residual extrapolation and information sharing across different stores for store category-level forecasting	2330 category–store time series from 336 stores, lengths range from 37 to 60 months	Marketing mixes and calendar effects	Leads 1–12 months; rolling out from 5 origins with the test data consisting of the following 12 months for each origin; MAE, MdAPE, MAPE	ADL, mixed model with AR(1), and Winters' ES	Residual extrapolation with information sharing across different stores helped to improve the accuracy for category-store forecasting.
Ma et al. (2016)	SKU-level retail store sales forecasting considering inter- and intra-category promotional effects	15 food categories, 926 SKUs from 1 grocery store, 320 weeks	Sales lags, price, features, displays, calendar events, week dummies	Rolling out-of-sample over 80 forecast origins, leads 1 and 4 weeks; MAE, RMSE, MASE, MPE, AvgRelMAE	Causal model only including focus SKU variables; univariate ETS model and base-lift model	The inclusion of competitive variables improved the forecasting accuracy. The LASSO scheme for model simplification was best.
Pillo et al. (2016)	Application of learning machines for sales forecasting under promotions	2 brands of pasta from 2 different retail stores, 3 years of daily sales	Calendar attributes, promotions, open hours, price, overall number of receipts	Divided last year of sample into 10 test intervals of the same size; rolling for 10 origins; MAPE	Univariate models: ARIMA, ES and HW	Both SVM and ANNs performed better than the univariate methods in the case of a suitable input selection.
Seeger et al. (2016)	Large-scale intermittent demand forecasting	Two intermittent databases considered: automobile spare parts with 20K items, and Amazon SKUs, 40K and 535K	Out-of-stock conditions and calendar effects, price changes	Point predictions over various lead times; quantiles	ETS; Snyder et al. state space model for intermittent demand (negative binomial)	Generalized linear model designed for intermittent data and including price and promotional effects, seasonality and holidays. Outperformed basic ETS.
da Veiga et al. (2016)	Applicability of natural computing approaches in foodstuff retail demand forecasting	63 liquid dairy products monthly sales in 3 product group, 108 months, aggregated at national level	None	Horizon 1–12 months;12 months of holdout; MAPE and U-Theil	ARIMA and HW	Wavelets neural networks provided better forecasts than the benchmarks.
Flunkert et al. (2017)	New development of applying neural nets to intermittent data in large scale applications	Various intermittent data bases considered: together with Amazon SKUs, 40K and 535K: 52 weeks, evaluated on the next year	Product category, seasonality; price was considered but added little	Horizons 1, 1–9; point predictions over various lead times; quantiles, cumulative demand	Snyder et al.'s state space model for intermittent demand	AR recurrent NN outperformed both the baseline and Seeger's (2016) approach.
Ramos and Fildes (2017)	Comparative study on univariate and multivariate forecasting methods on store × SKU data	988 SKUs in 203 categories: intermittent demand excluded, 173 weeks	Seasonal data, holidays, promotional events	Horizon 1 week ahead; rolling origin calculation with updated model parameters; many error measures included MAPE, RelMAE etc.	ETS, TBATS, ARIMA, Naïve: compared to multivariate alternatives and Lasso	Multivariate model including promotions performed best, with TBATS being the best of the univariate methods considered.

Abbreviation of forecasting models in the table: ADL: autoregressive distributed lag; ARIMA: autoregressive integrated moving average; MARMA: multivariate ARMA models; BVAR: Bayesian vector autoregression; ETS: error-trend-seasonal state space formulations of exponential smoothing; HW: Holt–Winters' procedure; MARS: multivariable adaptive regression splines; MLPNN: multi-layered perceptron neural network; MLR: multiple linear regression; RVM: relevance vector machine; SARIMA: seasonal autoregressive integrated moving average; SARIMAX: seasonal autoregressive integrated moving average with external variables; SES: simple exponential smoothing; SVR: support vector regression; TBATS: trigonometric, Box-Cox transform, ARMA errors, trend and seasonal components.

Abbreviation of accuracy metrics in the table: ME: mean error; MAD: mean absolute deviation; MPE: mean percentage error; MdAPE: median absolute percentage error; RMSE: root mean square error; ARMSE: average root mean squared error; MAPE: mean absolute percentage error; RMSPE: root mean square percentage error; MASE: mean absolute scaled error; sMAPE: symmetric mean absolute percentage error; AvgRelMAE: average relative mean absolute error; MSFE: mean squared forecast error; FESD: forecast error standard deviation.

A.2. Retail demand planning vignettes

See Table A.2.

Table A.2

	Case 1	Case 2	Case 3
Primary focus	Household, FMCG and garden	Food	Fashion
Product range	20K regular products, 20K new or short-season.	26K	20K, of which 70% regular products with sales history. Final forecasts based on SKUs × size profiles. *Variation: 100K, 80% daily intermittency at the DC level.*
Data characteristics	2 years of data: high intermittency (around 70%), promotional intense (10 types). Some 10% of regular SKUs are promoted weekly for a period of 4–6 weeks.	5 years of data used. 40% intermittent. 45% promoted within the last 40 days.	For regular products, one year of history. 70% intermittent at a store × day level.
Distribution	450+ primarily high-street stores, UK, with 2 distribution centres, plus online.	4 distributions centres plus online.	6 DCs, 350 stores + online.
Software	Demand forecasts at the store level provided by SAP's F&R (forecasting and replenishment) module. The F&R module is regression based and used dummy variables for the promotional types, seasonality, holidays and other events. *Variation: Aggregated store forecasts used to produce DC forecasts.*	SAS data handling with a tailored regression-based forecasting module: the model takes seasonal (day, month) effects, promotions including shelf space, weather and events. *Variation: AI-based software for stores, Relex (extrapolative) software at DC.*	JDA demand and fulfillment.
Forecasting (established products)	Weekly buckets with a 22+ week lead time, pro-rated to daily with forecasts updated daily for retail site ordering. Promotional forecasts 6+ weeks ahead based on an agreed promotional plan. *Variation: promotional forecasts done 'manually'.*	These forecasts take into account the day of the week, monthly effects, promotional effects including additional space, weather and events. Forecasts can be overridden by users.	Forecasts weekly (SKU × store), disaggregated by daily profile. Automatic univariate exponential smoothing based forecast with event dummies. Promotions forecast outside system. Lost sales calculation. Clustering of SKUs for calculating seasonal profiles.
Forecasting (new products)	Judgment supplemented by analogy.	Judgment supplemented by analogy (from the sales history of similar products). Demand planners will then react as demand is observed to ensure compatibility with the forecasts.	Judgmental estimates influenced (though not formally) by previous analogous SKUs. Seasonal profile for short-life products.
Forecasting (online sales)	Online products are forecast separately and are not linked to store sales.	Online sales are merged with local stores. *Variation: outsourced. No cannibalization assumed.*	Total SKU sales forecast with a hierarchical forecast by channel.
Inventory data: quality? Used in demand forecasting?			
Demand planning staffing	Approximately 8 FTEs plus 2 analysts with general supply chain planning and forecasting responsibilities. DC forecasting for logistics team. *Variation: supported by a data science team.*	Category-based (with 8 demand planners and 3 supply planners). Central forecasting team of 13 responsible for the smooth operation of system forecasts.	A central analytical forecasting team of 4, with planning managers in the different business units responsible for the different categories.
Collaboration with suppliers	None. *Variation: Order forecast for promotional items shard for delivery to DC. Inventory information shared with strategic suppliers.*	Information shared on order forecasts based on demand forecasts.	
Interventions	F&R is designed to handle multiple promotional types but omits other potentially important factors such as weather and intra-category effects. Approximately 5% of system forecasts are adjusted.	Interventions primarily across all stores daily. In addition, individual store adjustments are made too, leading to 40% of forecasts being affected. *Variation: 99% of forecasts affected.*	Limited adjustment of system forecasts for regular products – only by exception.
KPIs	Accuracy (measured by the MAPE per SKU across stores) 1 week ahead only, despite a focus on the next 6 weeks. *Variants: out-of-stock and inventory.*	Bias prioritized at the store level. *Variation: Sqrt(Sigma(A^2)/Sigma(F^2)).*	Out-of-stocks, forecast accuracy, stock level, reach.
Key challenges	New product forecasting and short life cycle; weather effects, variants: promotional effects, intermittent items.	Weather effects. Stock-outs. Identifying changes in dynamic retail market. Unique events. *Variation: hourly forecasts needed.*	Daily effects around holidays, monthly effects, new listings, weather, performance/runtime. Potential benefits of ML methods?

(*continued on next page*)

Table A.2 (*continued*).

	Case 4	Case 5
Primary focus	Beauty, body care, healthcare	Convenience retail, foodservice
Product range	15K regular products; 55M product-locations worldwide.	2k regular products, 500 intermittent/seasonal.
Data characteristics	110 weeks of time series.	110 weeks of sales history, 5 years of data available in data cubes. 10% of articles are on promotion more than 25% of the year, and an additional 15% are promoted at least twice a year. Promotions typically run 4-6 weeks at a time irregularly.
Distribution	About 3600 stores and 25 DCs internationally, online.	650 stores, 3 Mid-Atlantic DCs; 150 stores, 2 Florida DCs; select forecasting for 3 small DSD companies.
Software	SAP F&R for store forecasting and replenishment and DC forecasting, SAP retail for DC replenishment. Some custom development.	SAP F&R (forecasting and replenishment) module including promotions, holidays and weather events.
Forecasting (established products)	Weekly forecast with daily profiles. No promotions, EDLP strategy.	Weekly forecast going out 13 weeks (26 weeks for about 100 stores); weekly forecasts are disaggregated to daily buckets via dayweights.
Forecasting (new products)	No forecast in the first week, then new listing algorithms with special safety stock calculation. For relaunches, use sales reference logic.	New articles are referenced to existing items based on marketing department expectations of sales and patterns.
Forecasting (online sales)	Use product movements out of the online fulfilment DC as a basis for the weekly forecast.	N/A
Inventory data: quality? Used in demand forecasting?	Good data quality because of QA mechanisms.	Quality is intermittent; inventory is not taken into account for products with short shelf lives.
Demand planning staffing	14 FTEs for store forecasting and replenishment.	3 FTE demand planners, 1 FTE replenishment analyst. Also 1 system admin and 3 associates focused on store-level issues including opening/closing new stores and remodels.
Collaboration with suppliers	Yes	6-week order forecasts are provided to external partners. The marketing department provides longer-range expectations to key suppliers.
Interventions	Store- and category-specific interventions through custom development, e.g., monthly effects, holidays (and surrounding days), sometimes new listings (if conspicuous).	Correctional judgmental adjustments are applied for first-time occurrences. The data are corrected for weather events. Approximately 75 stores are considered "highly seasonal" and receive extra attention/profile adjustments during the summer season.
KPIs	Accuracy (measured by WMAPE by SKU across stores), store order acceptance, in-stock conditions; do not measure overstocks.	Forecast accuracy, measured by MAPE; out-of-stock occurrences and days of inventory.
Key challenges	Short shelf life product replenishment. Accurate dayweighting for fresh items. Weather effects. Data quality. Lack of reporting/root cause analysis.	Promotional effects; intermittency; inventory data accuracy and vendor fill rate data.

Notes: WMAPE weights the MAPE of SKUs across stores according to a volume (or value) measure.

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